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Decodificarea neurală a percepţiei vizuale din fMRI folosind modele generative ghidate de CLIP

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DIPLOMA THESIS

Neural Decoding of Visual Perception from fMRI Using CLIP-Guided Generative Models

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Abstract

This bachelor thesis studies neural decoding of visual perception from functional magnetic resonance imaging (fMRI) by learning a mapping from brain activity to CLIP-aligned image representations and by using those representations for retrieval and downstream image reconstruction. The work is developed in the setting of the Natural Scenes Dataset, where a subject observes natural images while 7T fMRI activity is recorded. Instead of predicting pixels directly, the proposed pipeline predicts image-aligned latent variables that are more robust for contrastive learning, nearest-neighbour retrieval, and controlled qualitative generation.

Chapter 1 introduces the problem, objectives, original contributions, and the transparency statement on limited AI-assisted writing. Chapter 2 reviews the theoretical background, the historical evolution of visual decoding, and the most relevant literature. Chapter 3 presents preprocessing, latent targets, losses, architecture design, and the evolution from compact–rerank decomposition to the final frozen fusion recipe. Chapter 4 documents the software design, validation strategy, and reproducibility measures of the implemented system. Chapter 5 reports the experimental trajectory, the final results, qualitative retrieval evidence, and the main limitations, while Chapter 6 concludes the thesis and outlines future work.

The main original contribution is a reproducible retrieval-oriented framework in which compact retrieval, dedicated reranking, rich latent regression, and later cross-architecture expert fusion are treated as distinct but cooperative components. The V30/V35 wave established the compact, rerank, and rich decomposition as the methodological foundation of the project, with the V35 plus frozen legacy N1v28a fixed normalized-weighted fusion reaching 77.6% validation Recall@1 and 77.2% SHARED1000 Recall@1 as an important milestone. The final exported system extends this foundation with a frozen cross-architecture triple score fusion that combines the V61a 197K token-space model with 16-draw MC-TTA, the V62a 768-D CLS model, and the V66a 768-D ROI-pretrained model under fixed CSLS-based score weights, reaching 86.3% CSLS Recall@1 on SHARED1000 as the paper-grade endpoint. The qualitative diffusion add-on is interpreted conservatively: a curated 16-example evaluation anchored on the final triple-fusion top-1 retrievals, conditioned on the fMRI-predicted CLIP embedding with best-of-16 candidate selection and uncertainty-aware img2img strength, exposes a trade-off between low-level fidelity and semantic steering rather than a uniform improvement, so retrieval remains the primary quantitative contribution of the thesis.

Keywords: fMRI, neural decoding, CLIP, diffusion models, retrieval, NSD.

Contents

Chapter 1

Introduction

Recovering visual information from brain activity is one of the most ambitious goals of modern computational neuroscience. In recent years, large-scale neuroimaging datasets, multimodal representation learning, and generative models have enabled a shift from coarse category decoding toward image-specific retrieval and plausible visual reconstruction from fMRI signals [1, 9, 12, 13]. This progress is scientifically important because it connects two difficult problems: understanding how the visual cortex represents semantic and structural information, and learning machine representations that can act as bridges between biological measurements and natural images.

The present thesis focuses on the *visual perception* setting. During each trial, a subject observes a natural image and the corresponding blood-oxygen-level-dependent (BOLD) response is recorded with high-field fMRI. The central task is to infer an image-aligned latent representation from this neural signal. Rather than predicting pixels directly, the pipeline maps fMRI activity into a CLIP-based embedding space, where similarity can be measured reliably and where downstream diffusion-based image generation becomes possible [10, 11]. This choice turns the problem into a structured latent prediction task rather than a fragile direct image regression problem.

Formally, let $x_i \in \mathbb{R}^V$ denote the preprocessed voxel vector for trial i, where V is the number of selected voxels, and let $z_i \in \mathbb{R}^d$ denote the target image embedding extracted from a pretrained vision-language model. The decoder is a parametric mapping

$$\hat{z}_i = f_\theta(x_i), \qquad f_\theta : \mathbb{R}^V \to \mathbb{R}^d, \tag{1.1}$$

where $\hat{z}_i$ should preserve enough image-specific information to retrieve the correct image from a held-out gallery and, later, to condition a generative model. The main evaluation scenario is therefore retrieval: for each predicted embedding, the system searches a gallery of candidate images and ideally ranks the ground-truth item first.

This formulation is challenging for several reasons. First, fMRI is an indirect and noisy measurement of neural activity, with low temporal resolution and substantial inter-session variability [1, 20]. Second, the representation space of modern visual models is high-dimensional and geometrically nontrivial [10, 3]. Third, there is a constant risk of data leakage or protocol mismatch when repeated images, subject-specific preprocessing, and cached features are involved. A strong bachelor thesis in this area must therefore be both algorithmically ambitious and experimentally rigorous.

The main research question of this thesis is the following: *how can one learn a stable mapping from cortical activity to an image-aligned latent space that supports accurate retrieval and provides a credible foundation for visual reconstruction?* The project approaches this question through an iterative experimental program. Early stages establish robust baselines and eliminate implementation errors. Later stages explore probabilistic decoders, directional losses, richer targets, multi-head retrieval designs, and shortlist-based fusion strategies.

1.1 Motivation and Context

The work is motivated by the convergence of three research directions. The first is the availability of datasets such as the Natural Scenes Dataset, which provide image-level behavioural and neuroimaging data at a scale that was previously unavailable [1]. The second is the emergence of CLIP-style multimodal latent spaces, which align image representations with semantic structure and perform well in retrieval settings [10]. The third is the success of latent diffusion models, which make it possible to turn recovered latent information into realistic image samples [11, 9, 16]. Together, these directions make fMRI-to-image decoding both more realistic and more measurable than in earlier generations of brain-decoding systems.

A complementary single-example view is useful because it makes the retrieval-first logic concrete. Starting from one viewed stimulus and its matched cortical response, the thesis builds a CLIP target embedding from the image, learns a brain-to-latent mapping from the associated fMRI activity, and then evaluates the predicted latent first through held-out retrieval and only secondarily through optional qualitative generation.

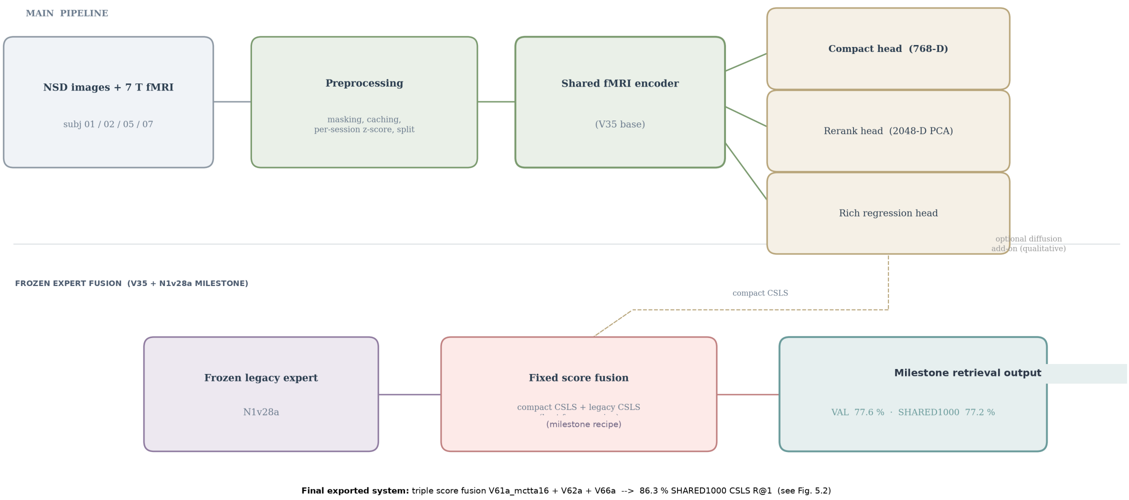


Figure 1.1: Overview of the complete thesis pipeline. The written thesis is centered on retrieval: NSD fMRI is preprocessed and mapped by a shared encoder into compact, rerank, and rich latent heads. The V35 plus frozen legacy N1v28a fixed fusion is retained as a methodological milestone, while the final exported system is a frozen cross-architecture triple score fusion that combines a 197K token-space model with MC-TTA, a 768-D CLS model, and a 768-D ROI-pretrained model. The diffusion add-on is treated as an optional qualitative extension rather than as the primary benchmark.

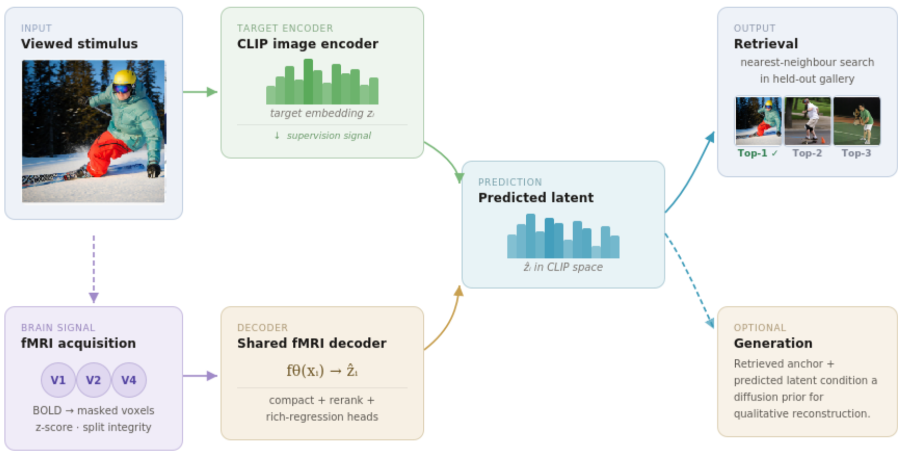


Figure 1.2: Single-example decoding path used throughout the thesis. A viewed image defines the target CLIP embedding, the corresponding fMRI response is preprocessed into a voxel vector, and the decoder learns to map that brain activity into the shared latent space used for retrieval-first evaluation and optional downstream generation.

1.2 Objectives of the Thesis

The thesis has four concrete objectives:

- to construct a reproducible pipeline for mapping fMRI activity to CLIP-aligned visual embeddings;
- to evaluate how architectural and loss-design choices affect retrieval quality under a leakage-free protocol;
- to analyse why some experimentally attractive ideas fail, especially under hubness and representation-geometry constraints;
- to validate a multi-stage retrieval strategy in which compact retrieval, dedicated reranking, and frozen score-level fusion across complementary experts are treated as distinct but cooperative inference mechanisms.

These objectives are intentionally both scientific and engineering-oriented. The final artifact is not only a set of metric values, but also a structured experimental framework in which preprocessing, target caching, training, evaluation, qualitative analysis, and implementation validation are integrated into a single research pipeline.

1.3 Original Contributions of the Author

The original contributions emphasized in this thesis are the following:

- the design and implementation of an end-to-end fMRI-to-latent decoding pipeline built around NSD, CLIP targets, and retrieval-based evaluation;
- a systematic debugging and validation program addressing alignment, split integrity, checkpoint selection, and retrieval geometry;
- an experimental comparison between single-head, dual-head, and triple-head retrieval formulations under a shared protocol;
- the validation of shortlist-local reranking as a corrective mechanism, followed by a stronger frozen fixed-fusion milestone built on the V35 compact expert and the legacy N1v28a expert, and finally extended into a frozen cross-architecture triple score fusion that combines a 197K token-space model with MC-TTA and two complementary 768-D experts as the final exported retrieval system;

- a conservative qualitative analysis of retrieval-guided diffusion refinement using SD 2.1 img2img with CLIP-conditioned best-of-N selection and uncertainty-aware strength, which exposes a low-level-versus-semantic trade-off rather than a uniform improvement over the retrieval anchor;
- the explicit documentation of negative results and failed directions, which is scientifically useful because it narrows future search space.

At the time of writing, the work has not yet been presented as a peer-reviewed conference publication. The thesis therefore serves as the primary written record of the experimental results, methodological design, and implementation choices.

1.4 Transparency Regarding the Use of Generative AI Tools

In accordance with the university guidelines, the author states transparently that generative AI tools were used only in a limited and assistive manner during the preparation of the manuscript. Their use was restricted to language polishing, reformulation suggestions, and LaTeX refactoring support. The scientific content, literature verification, experimental design, software implementation, numerical results, and interpretation of findings remain the full responsibility of the author. No generative AI system is treated as an author or as an authoritative source of scientific truth.

1.5 Structure of the Thesis

The remainder of the thesis is organized as follows. Chapter 2 reviews the theoretical background, the relevant literature, the historical evolution of the field, the NSD dataset, and the main evaluation metrics used in the project. Chapter 3 presents the proposed methodology, including preprocessing, target construction, model design, and the multi-head retrieval formulation. Chapter 4 describes the software design, experiment management, validation strategy, and reproducibility measures of the implemented system. Chapter 5 reports the experimental evolution of the project, summarizes the main results, and discusses their scientific interpretation. Finally, Chapter 6 summarizes the conclusions, limitations, and future directions.

Chapter 2

Theoretical Background and Related Work

2.1 Domain and Subdomain of the Thesis

The thesis belongs to the broad domain of artificial intelligence applied to computational neuroscience. More specifically, it is situated at the intersection of brain decoding, representation learning, and generative modelling. The subdomain addressed here is *visual perception decoding from fMRI*: given a measured cortical response, the goal is to recover information about the external image stimulus that produced it. This task differs from classical neuroimaging analysis because it is not limited to identifying which brain regions are active; instead, it seeks to infer *what was seen* by mapping neural activity into a meaningful computational representation [6, 7].

2.2 Historical Evolution of Visual Decoding

The modern history of visual decoding from fMRI can be understood as a progression from simple stimulus recovery to semantically rich latent reconstruction. Early work demonstrated that activation patterns in the visual cortex contain sufficient information to infer low-level stimulus structure. Thirion et al. proposed *inverse retinotopy*, showing that it is possible to infer the visual content of images from retinotopically organized activation patterns [17]. Shortly afterward, Kay et al. showed that natural images can be identified from fMRI responses in visual cortex, marking an important step from simple patterns toward naturalistic stimuli [6]. Naselaris et al. further introduced Bayesian reconstruction of natural images from brain activity, showing that explicit generative modelling can improve visual recovery [7].

The next milestone was the transition from static natural images to dynamic natural vision. Nishimoto et al. reconstructed viewed movie clips from fMRI responses evoked by natural movies, demonstrating that the decoding framework can generalize beyond isolated stimuli [8]. Later work increasingly relied on machine-vision features and deep networks. Horikawa and Kamitani showed that hierarchical visual features decoded from fMRI can identify both seen and imagined objects [5]. Wen et al. extended deep encoding/decoding ideas to dynamic natural vision [20]. Generative-model-based approaches then became prominent: Seeliger et al. used GANs for natural-image reconstruction [14], Shen et al. combined decoded multilayer DNN features with deep image reconstruction [15], and VanRullen and Reddy reconstructed faces using deep generative neural networks [19].

Recent work has moved decisively toward multimodal latent spaces and powerful generative priors. Takagi and Nishimoto used latent diffusion to reconstruct images from brain activity [16], Ozcelik and VanRullen proposed generative latent diffusion for natural-scene reconstruction [9], MindEye introduced a modern retrieval-plus-reconstruction framework [12], and MindEye2 showed that shared-subject models can dramatically reduce the per-subject data requirement [13]. The present thesis belongs to this recent phase, with a strong emphasis on retrieval-oriented latent decoding.

2.3 fMRI-Based Visual Decoding

Functional magnetic resonance imaging measures the BOLD signal, which acts as an indirect proxy for neural activity [1, 20]. In visual decoding tasks, the input is usually a voxel vector derived from one or more regions of interest in the visual cortex, while the target corresponds to some representation of the viewed stimulus. Earlier approaches focused on category labels or hand-crafted image descriptors [17, 6, 7]. More recent work predicts features from deep visual models, which allows the decoder to preserve semantic and structural information at a much richer level [5, 20, 12].

Let $x_i \in \mathbb{R}^V$ be the voxel vector for trial i. A basic decoder predicts $\hat{z}_i \in \mathbb{R}^d$, where z_i is the target visual representation extracted from the ground-truth image. Because the embedding space is usually normalized, cosine similarity becomes the natural scoring function:

$$\cos(\hat{z}_i, z_j) = \frac{\hat{z}_i^\top z_j}{\|\hat{z}_i\|_2 \, \|z_j\|_2}. \tag{2.1}$$

A retrieval system ranks all gallery items according to this similarity and checks whether the correct image is recovered near the top of the list.

Year	Representative work	Main contribution	Relation to the present thesis
2006	Thirion et al. [17]	Inverse retinotopy and low-level visual inference	Historical starting point for visual-content recovery
2008–2009	Kay et al. [6]; Naselaris et al. [7]	Natural-image identification and Bayesian reconstruction	Early foundations for natural-image decoding
2011	Nishimoto et al. [8]	Reconstruction from natural movie responses	Demonstrates naturalistic decoding feasibility
2017–2019	Horikawa and Kamitani [5]; Shen et al. [15]; VanRullen and Reddy [19]	Deep-feature decoding and early modern reconstruction approaches	Bridges classical decoding and contemporary latent modelling
2023–2024	Takagi and Nishimoto [16]; Ozcelik and VanRullen [9]; Scotti et al. [12, 13]	Diffusion priors, CLIP-like targets, shared-subject scaling	Closest literature context for this thesis

Table 2.1: Short historical timeline of visual decoding from brain activity.

2.4 Natural Scenes Dataset

The Natural Scenes Dataset is one of the most important benchmarks for high-resolution visual decoding from fMRI [1]. It contains 7T fMRI responses recorded while subjects viewed thousands of natural images. The scale, signal quality, and repeated-image structure of NSD make it especially suitable for studying image-specific decoding. However, NSD also imposes strict requirements on data handling: repeated trials, image identifiers, train/validation separation, and session-level normalization must all be treated carefully to avoid leakage.

In this thesis, the practical role of NSD is twofold. First, it provides the paired (x_i, z_i) examples needed for supervised learning. Second, it offers a benchmark environment in which the quality of retrieval and reconstruction can be discussed relative to recent systems such as MindEye and MindEye2 [12, 13]. The present work does not attempt to replace NSD-specific discipline with ad hoc experimentation; on the contrary, it treats split integrity and reproducibility as first-class methodological requirements.

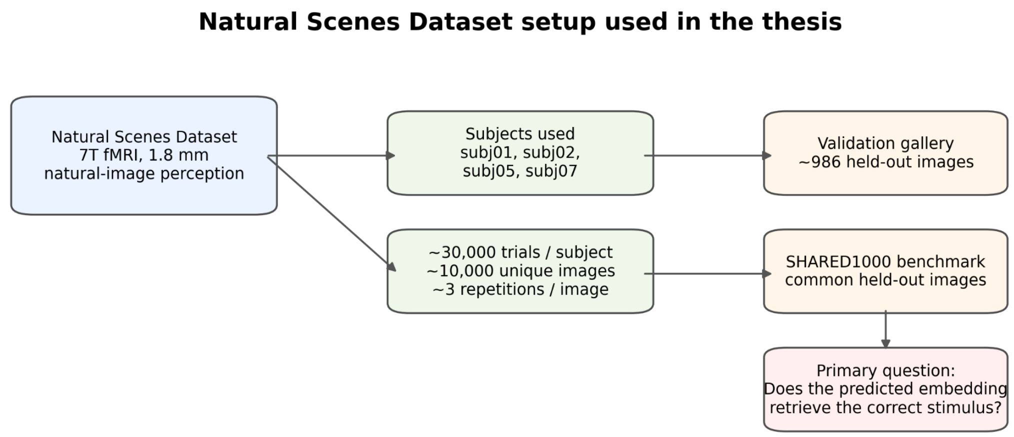


Figure 2.1: Dataset setting used throughout the thesis. The work is developed on the Natural Scenes Dataset under a retrieval-oriented protocol, with validation and SHARED1000 treated as the main endpoints.

2.5 CLIP as the Target Representation

CLIP learns a shared representation space for images and text using large-scale contrastive supervision [10]. Its image encoder produces embeddings that are semantically meaningful, well structured for nearest-neighbour search, and highly reusable across downstream tasks. For brain decoding, CLIP is useful because it provides a target space that is already organized around visual-semantic similarity.

This thesis uses CLIP-style image embeddings as latent targets instead of direct pixel supervision. The practical benefit is that the decoder does not need to reconstruct raw images directly from noisy brain signals. Instead, it predicts a representation that can support both retrieval and conditioning of a downstream generator. This idea aligns well with recent diffusion-based reconstruction pipelines [16, 9, 12].

Figure 2.2 illustrates the geometric intuition behind this choice. CLIP places semantically corresponding images and text descriptions in neighboring regions of a shared embedding space. In the present thesis, this shared geometry becomes the decoding target: a predicted brain embedding is useful only insofar as it lands near the correct image neighborhood while remaining separated from semantically plausible distractors.

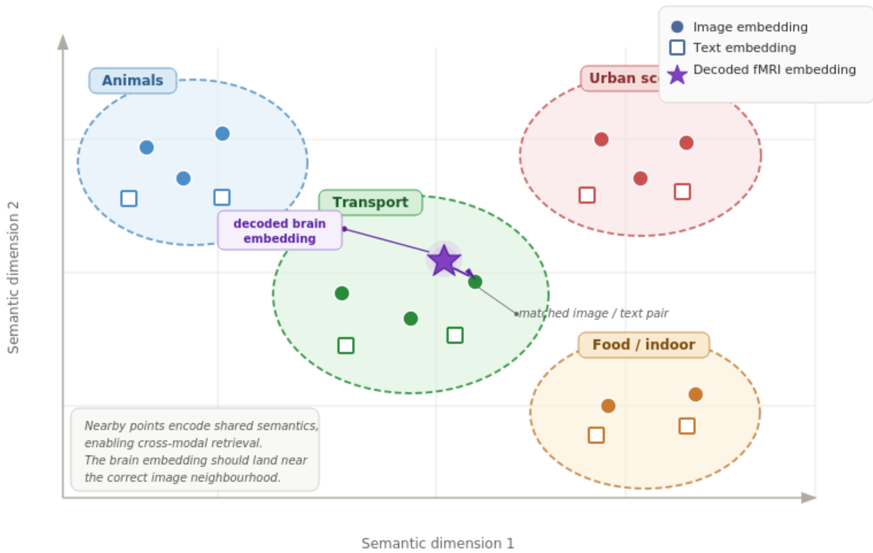


Figure 2.2: Conceptual view of CLIP as a shared semantic space. Image and text embeddings for related concepts occupy nearby regions, which is why CLIP provides a meaningful target geometry for brain decoding and retrieval.

2.6 Latent Diffusion Models and Generative Reconstruction

Latent diffusion models synthesize images by iteratively denoising a latent variable conditioned on text, image, or other contextual information [11]. In the context of brain decoding, the role of a diffusion model is typically not to replace retrieval, but to transform recovered latent information into a visual reconstruction. If the predicted brain-derived embedding retains enough semantic and structural content, a diffusion prior can use that information to generate an image that resembles the viewed stimulus [16, 9].

At a high level, a forward diffusion step perturbs a clean latent y_0 into a noisy variable y_t according to

$$y_t \sim \mathcal{N}\Big(\sqrt{1-\beta_t}\, y_{t-1},\ \beta_t I\Big)\,. \tag{2.2}$$

and the learned reverse process attempts to recover a clean latent conditioned on the predicted visual representation. In the present thesis, reconstruction is treated as an important downstream extension, while retrieval remains the main quantitative focus.

2.7 Retrieval Geometry and Hubness

A central conceptual issue in this thesis is that accurate latent regression is not sufficient by itself; what matters quantitatively is the geometry of the predicted embeddings relative to the gallery. High-dimensional retrieval spaces often suffer from *hubness*, meaning that a few gallery items become nearest neighbours for an excessive number of queries. This phenomenon can distort raw cosine retrieval and make a model appear less discriminative than its pointwise regression error would suggest [3].

This is why the project relies not only on raw similarity but also on hubness-aware scoring. The distinction between absolute latent fidelity and retrieval geometry became one of the key interpretive lessons of the experiments. It also explains why some seemingly attractive modelling changes improved regression behaviour without improving the final retrieval metric.

2.8 Evaluation Metrics

The core metric used throughout the thesis is Recall@K. For a predicted embedding $\hat{z}_i$ and a gallery $\mathcal{G}$, Recall@K is defined as

$$\mathrm{R@K} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\mathrm{rank}(g_i^\star \mid \hat{z}_i) \leq K\right], \tag{2.3}$$

where $g_i^\star$ is the ground-truth gallery image corresponding to sample i. In this thesis, Recall@1 is the most important endpoint because it directly measures image-specific identification.

A second important metric is cross-domain similarity local scaling (CSLS), which corrects for hubness in high-dimensional retrieval spaces [3]. Let $r_q(\hat{z}_i)$ denote the average similarity between query $\hat{z}_i$ and its k nearest gallery neighbours, and let $r_g(z_j)$ denote the average similarity between gallery item z_j and its k nearest query neighbours. Then CSLS is defined as

$$\mathrm{CSLS}(\hat{z}_i, z_j) = 2\,\cos(\hat{z}_i, z_j) - r_q(\hat{z}_i) - r_g(z_j). \tag{2.4}$$

CSLS is particularly relevant in the present project because the experiments repeatedly show that raw regression fidelity and retrieval geometry are not the same thing.

For the reconstruction add-on the thesis additionally reports image-fidelity scores that have become standard in the fMRI-to-image literature [12, 13, 9]: pixel correlation, SSIM, PSNR, LPIPS, CLIP image similarity, and feature-space cosine similarity at two depths of an ImageNet-pretrained AlexNet. Following the convention used by MindEye and Brain Diffuser, AlexNet(2) denotes cosine similarity between the activations of the second convolutional block (early, mostly texture/edge features), and AlexNet(5) denotes cosine similarity between the activations of the fifth convolutional block (mid-level, more semantically organized features). All AlexNet activations are flattened and ℓ_2-normalized before the cosine product so that the metric is scale-invariant. These metrics are auxiliary, not retrieval endpoints, and are reported only over the curated 16-example reconstruction subset rather than over SHARED1000 as a whole.

2.9 Representative Related Work

Recent work on fMRI-to-image decoding has shifted decisively toward learned latent representations. Ozcelik and VanRullen used latent diffusion conditioned on brain-derived features to reconstruct viewed images from fMRI, demonstrating that generative priors can compensate for the noisiness of neural measurements [9]. Takagi and Nishimoto similarly showed that large pretrained diffusion and language models can be adapted for high-quality reconstruction from human brain activity [16]. MindEye and MindEye2 extended this direction with stronger contrastive objectives, richer target representations, and shared-subject learning, pushing the field toward more reliable image-specific decoding [12, 13].

From the perspective of this thesis, the most relevant lessons from the literature are the following. First, target representation quality matters greatly: richer embeddings often outperform simplistic regression targets. Second, data scale and subject alignment remain critical bottlenecks. Third, retrieval and generation should be treated as related but distinct objectives. These insights directly motivate the methodological design of the present work.

Work	Main idea	Representation / generator	Relevance to this thesis
Allen et al. (2022) [1]	Large-scale 7T benchmark dataset	NSD data resource	Provides the experimental foundation and evaluation setting
Radford et al. (2021) [10]	Vision-language contrastive pretraining	CLIP embedding space	Supplies the latent target space used for decoding
Takagi and Nishimoto (2023) [16]	Stable-Diffusion-based reconstruction	Latent diffusion conditioning	Shows how strong pretrained generators can be coupled with decoded features
Ozcelik and VanRullen (2023) [9]	Generative latent diffusion for natural scenes	Brain-to-latent diffusion conditioning	Motivates the reconstruction pathway after latent decoding
Scotti et al. (2023) [12]	Contrastive fMRI-to-image retrieval and reconstruction	CLIP-style targets and diffusion prior	Directly inspires the retrieval-oriented framing of the present work
Scotti et al. (2024) [13]	Shared-subject scaling for fMRI-to-image decoding	Shared-subject alignment and modern retrieval design	Serves as an upper benchmark and clarifies the role of scale and alignment

Table 2.2: Representative works most closely related to the thesis topic.

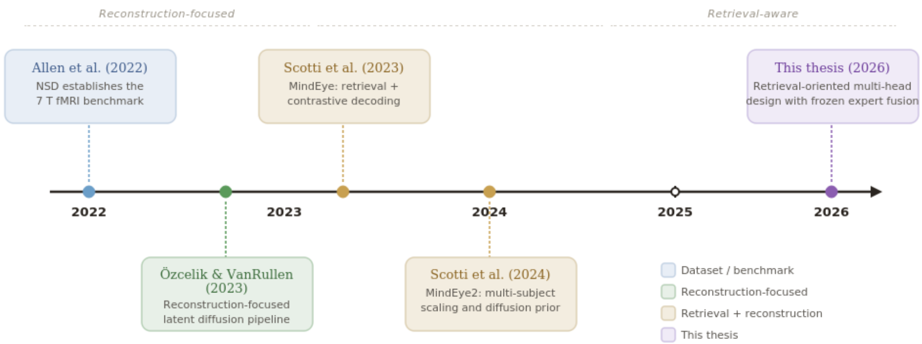


Figure 2.3: Positioning of the thesis relative to recent fMRI-to-image literature. The figure emphasizes the shift from early reconstruction-focused systems toward retrieval-aware and multi-subject systems, which provides the broader context for the retrieval-oriented design choices of this thesis.

Chapter 3

Proposed Methodology

3.1 Problem Formulation

The thesis addresses supervised neural decoding in the perception setting. Each training example consists of a pair (x_i, z_i), where x_i is the preprocessed fMRI feature vector and z_i is a target image representation extracted from the corresponding viewed stimulus. The learning objective is to estimate a decoder f_θ such that the predicted representation

$$\hat{z}_i = f_\theta(x_i) \tag{3.1}$$

is close to the correct target and sufficiently discriminative relative to all other targets in the gallery.

In practice, the decoder does not solve a single homogeneous problem. The later stages of the thesis distinguish between three complementary roles: compact retrieval, reranking, and rich regression. This leads naturally to a multi-head architecture,

$$\left(\hat{z}_i^{(c)}, \hat{z}_i^{(r)}, \hat{z}_i^{(g)}\right) = f_\theta(x_i), \tag{3.2}$$

where the superscripts denote the compact head, rerank head, and rich generative/regression head, respectively.

3.2 Data Preparation and Preprocessing

The preprocessing pipeline transforms raw NSD-derived activity into model-ready vectors. The core steps are voxel selection, repeated-trial aggregation when needed, subject/session indexing, per-session normalization, and alignment with image identifiers and cached visual targets. A crucial principle throughout the project is that train/validation separation must occur at the image level, not merely at the trial level, so that repeated observations of the same image do not leak across splits [1].

If $x_{i,s}$ denotes the feature vector for sample i in session s, z-scoring is performed using statistics estimated only from the training data of that session:

$$\tilde{x}_{i,s} = \frac{x_{i,s} - \mu_s^{\text{train}}}{\sigma_s^{\text{train}} + \varepsilon}. \tag{3.3}$$

This reduces session-specific drift while preserving discriminative variation relevant to decoding. In practical terms, the preprocessing stage is not a mere data-cleaning step; it is part of the scientific method, because even a strong architecture becomes meaningless if the split logic or target lookup is incorrect.

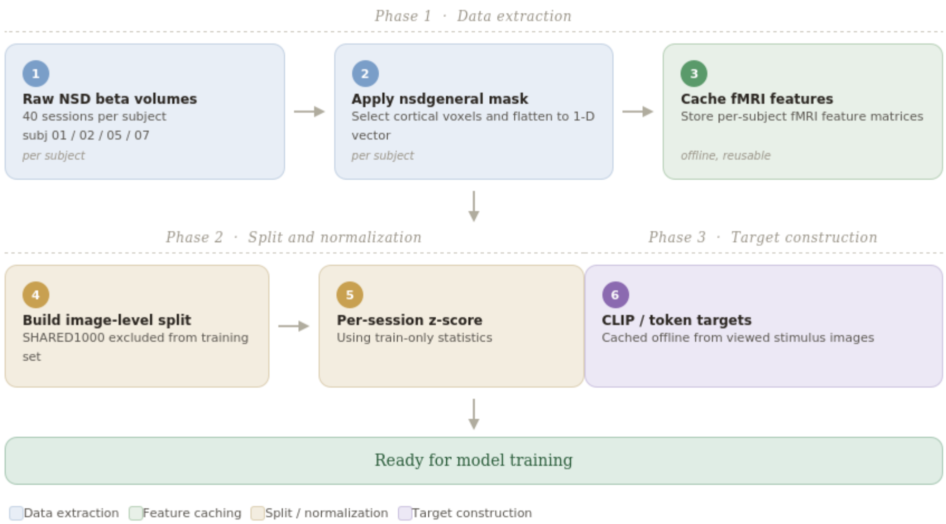


Figure 3.1: Preprocessing and target-construction workflow used in the thesis. The pipeline emphasizes offline caching, image-level split integrity, and per-session normalization before model training.

3.3 Latent Targets

The primary targets are CLIP image embeddings or derived token-level variants obtained from the viewed stimulus image. Let $\phi(I_i)$ denote the target representation extracted from image I_i by a pretrained visual model. The simplest pooled-target formulation uses

$$z_i = \phi(I_i) \in \mathbb{R}^d, \tag{3.4}$$

while richer variants preserve token-level or multi-scale structure. The experiments reported in the thesis show that target quality and granularity have a large effect on retrieval performance, especially when the decoder is strong enough to exploit that additional information [10, 12].

In the methodological evolution of the project, target design became one of the decisive axes of improvement. Compact pooled targets are easier to optimize for shortlist retrieval, whereas richer targets preserve more semantic and structural information for downstream regression and generative reconstruction. The final system therefore treats target richness as a design choice rather than as a universally monotone improvement.

3.4 Training Objectives

The compact head is primarily optimized for ranking. A standard contrastive formulation is the InfoNCE loss [18]:

$$\mathcal{L}_{\mathrm{NCE}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\cos(\hat{z}_i^{(c)}, z_i)/\tau\right)}{\sum_{j=1}^{N}\exp\left(\cos(\hat{z}_i^{(c)}, z_j)/\tau\right)}. \tag{3.5}$$

Here $\tau > 0$ is a temperature hyperparameter controlling the sharpness of the ranking distribution.

Some experiments model the predicted latent direction probabilistically through a von Mises–Fisher (vMF) output, a natural choice for unit-norm directional embeddings [2]. If μ_i is a unit-norm mean direction and κ_i is a concentration parameter, the density of the target direction can be written as

$$p(z_i \mid x_i) = C_d(\kappa_i)\exp\left(\kappa_i\,\mu_i^{\top} z_i\right), \tag{3.6}$$

where $C_d(\kappa_i)$ is the normalization constant on the unit hypersphere. This formulation is attractive because cosine geometry and directional uncertainty are naturally aligned; however, the thesis also shows that vMF training is sensitive to numerical pathologies if temperature and regularization are not tuned carefully.

The rich head is optimized with a regression-style objective that encourages absolute fidelity in latent space:

$$\mathcal{L}_{\mathrm{reg}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{z}_i^{(g)} - z_i\right\|_2^2. \tag{3.7}$$

The overall training objective combines multiple terms:

$$\mathcal{L}_{\mathrm{total}} = \lambda_c\mathcal{L}_{\mathrm{compact}} + \lambda_r\mathcal{L}_{\mathrm{rerank}} + \lambda_g\mathcal{L}_{\mathrm{reg}}, \tag{3.8}$$

where the λ coefficients control the relative importance of the different heads.

3.5 Architecture Evolution

The methodological evolution of the thesis begins with wide multilayer perceptron decoders operating on the full voxel vector. These models are intentionally simple and serve as strong baselines. Later iterations introduce probabilistic directional heads, ROI-aware tokenization, transformer-based components inspired by token processing in modern vision models [4], and finally multi-head retrieval architectures.

The final trainable architecture described in this thesis is the V30/V35 family, which separates the decoding problem into three subproblems:

- the **compact head**, which is optimized for shortlist retrieval in a stable low-dimensional space;
- the **rerank head**, which operates only on shortlist candidates and performs shortlist-local discrimination;
- the **rich head**, which preserves a higher-capacity representation useful for regression and reconstruction-oriented supervision.

This separation is one of the central design insights of the project. A single latent space is not forced to satisfy all design requirements simultaneously.

3.6 Inference and Score Fusion

At inference time, the compact head first retrieves a shortlist of candidate images. Let $\mathcal{S}_K(x_i)$ denote the top-K shortlist obtained from compact retrieval. In the earlier V30/V32 wave, the rerank head rescales those candidates locally. If $s_c(i,j)$ is the compact score and $s_r(i,j)$ is the rerank score for candidate j in the shortlist of sample i, the shortlist-local fused score can be written as

$$s_{\text{fusion}}(i,j) = \alpha\, s_c(i,j) + (1-\alpha)\, s_r(i,j), \qquad j \in \mathcal{S}_K(x_i), \tag{3.9}$$

where $\alpha \in [0,1]$ is a fusion weight. This formulation established the key methodological lesson of the thesis: reranking is most effective as a corrective signal applied inside a strong shortlist.

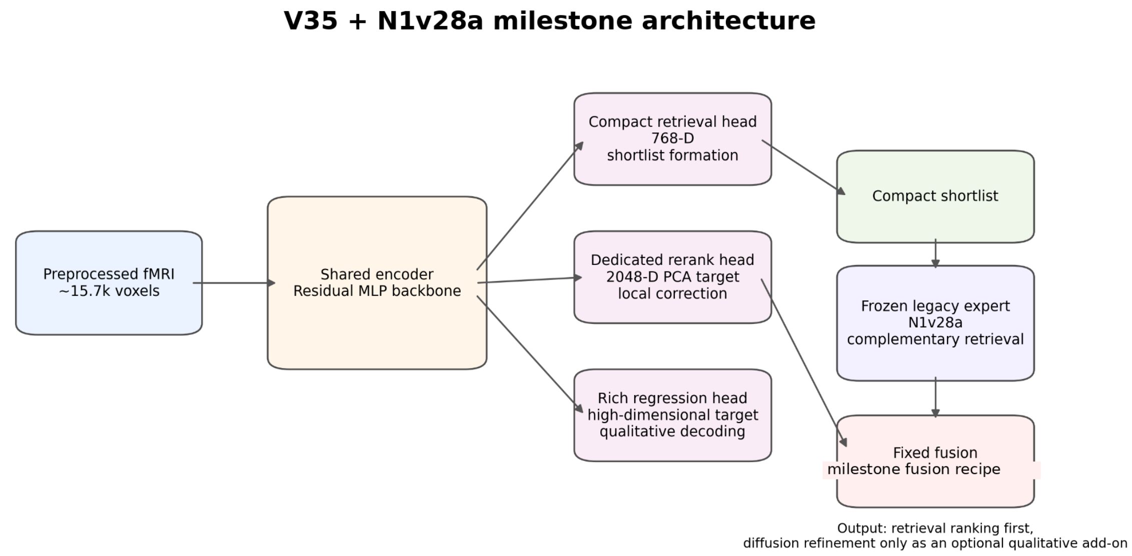


Figure 3.2: Decoder architecture corresponding to the V35 plus N1v28a fixed-fusion milestone. A shared encoder feeds a compact shortlist head, a dedicated rerank head, and a richer regression head, and the V35 compact expert is combined at inference time with the frozen legacy N1v28a expert. The final exported system retains the same multi-head decomposition and extends this expert-fusion idea further at inference time into the cross-architecture triple score fusion described in Section 5 (see Equation (3.11)).

A first frozen fixed-fusion system extends this idea with a complementary legacy expert. Using z-score-normalized expert scores, the corresponding score combination can be written as

$$s_{\text{milestone}}(i,j) = 0.3\,\widetilde{s}_{c,\text{CSLS}}(i,j) + 0.7\,\widetilde{s}_{\ell,\text{CSLS}}(i,j), \tag{3.10}$$

with $\widetilde{s}_{c,\text{CSLS}}$ denoting the normalized compact-CSLS score from V35 and $\widetilde{s}_{\ell,\text{CSLS}}$ denoting the normalized legacy-CSLS score from N1v28a. The rerank branch remains methodologically important and still informs the architecture; the V35 plus legacy combination is retained as a milestone result that validates the compact–rerank–rich decomposition under expert fusion.

After validating this compact, rerank, and rich decomposition, the final inference-time extension explored stronger token-space decoding and cross-architecture score fusion. The exported best system replaces the two-expert fusion above with a frozen score-level combination of three retrieval experts that span different target spaces and different inductive biases: a 197K token-space model (V61a) evaluated with 16-draw Monte Carlo test-time augmentation (MC-TTA-16), a 768-D CLS retrieval model (V62a), and a 768-D ROI-pretrained model (V66a). With $\widetilde{s}_{1,\text{CSLS}}$, $\widetilde{s}_{2,\text{CSLS}}$, and $\widetilde{s}_{3,\text{CSLS}}$ denoting the normalized CSLS scores of these three experts, the final retrieval score is

$$s_{\text{final}}(i, j) = 0.7\,\widetilde{s}_{1,\text{CSLS}}(i, j) + 0.1\,\widetilde{s}_{2,\text{CSLS}}(i, j) + 0.2\,\widetilde{s}_{3,\text{CSLS}}(i, j), \tag{3.11}$$

evaluated with CSLS neighbourhood size $k = 3$. The weights, the CSLS neighbourhood size, and the choice of experts are all frozen before SHARED1000 evaluation, so the final system performs no test-time tuning. It is important to emphasize that Equation (3.11) defines a score-level inference-time fusion rather than an end-to-end fusion-trained architecture, and the system does not produce a single unified fused latent embedding.

A useful way to interpret the overall design is therefore that the first stage solves a difficult global search problem, the rerank branch provides shortlist-local discrimination, the two-expert milestone fusion validates that complementary experts can be combined at inference time, and the final triple fusion combines partially different information from three target spaces and architecture families through frozen score-level weights, without retraining the backbone. This division of labour is one of the main methodological lessons of the project.

3.7 Reproducibility Principles

A significant part of the methodology is dedicated to reproducibility. The project maintains versioned experiment configurations, cached targets, explicit split logic, and a strict separation between training-time and evaluation-time computations. This is especially important in a thesis context because the scientific value of the results depends not only on the final metric values, but also on the ability to explain why those values were obtained and how they changed across experimental waves.

The practical implication is that every experiment is treated as a traceable artifact with explicit inputs, outputs, and evaluation scripts. In a research area where silent alignment bugs can invalidate an otherwise sophisticated system, reproducibility is not just a software-engineering virtue; it is a core scientific requirement.

Chapter 4

Software Design, Implementation, and Validation

4.1 Analysis and Design Goals

Because the thesis contains an implemented software system, the written document should reflect the essential stages of the application life cycle: analysis, design, implementation, and testing. In this project, the analysis stage identified four non-negotiable requirements. First, the system had to preserve strict alignment between each fMRI sample and its visual target. Second, it had to support rapid experimental iteration without silently changing the evaluation protocol. Third, it had to make it easy to compare model versions under a common metric suite. Fourth, it had to remain reproducible enough that every reported result could be traced back to a concrete configuration and checkpoint.

These requirements informed the high-level design of the pipeline. Instead of building a monolithic training script, the system was organized around loosely coupled stages: data preparation, target extraction and caching, model training, evaluation, result aggregation, and qualitative analysis. This decomposition reduces implementation risk and makes scientific debugging more transparent. If an experiment fails, the author can determine whether the failure is caused by preprocessing, target generation, optimization, or evaluation rather than treating the whole project as a black box.

4.2 Software Architecture

At a conceptual level, the implemented system follows a staged architecture:

1. **Data layer:** loading NSD-derived features, image identifiers, subject/session metadata, and split definitions;

2. **Target layer:** extracting, caching, and reusing CLIP-based latent targets and derived variants;

3. **Model layer:** defining baseline and multi-head decoders together with their losses and optimization logic;

4. **Evaluation layer:** computing retrieval metrics, shortlist reranking, fusion scores, and summary tables;

5. **Analysis layer:** storing experiment manifests, qualitative outputs, plots, and leaderboard-style comparisons.

This architecture is intentionally research-oriented. Its purpose is not only to train a model, but also to preserve experimental meaning and make repeated ablations comparable.

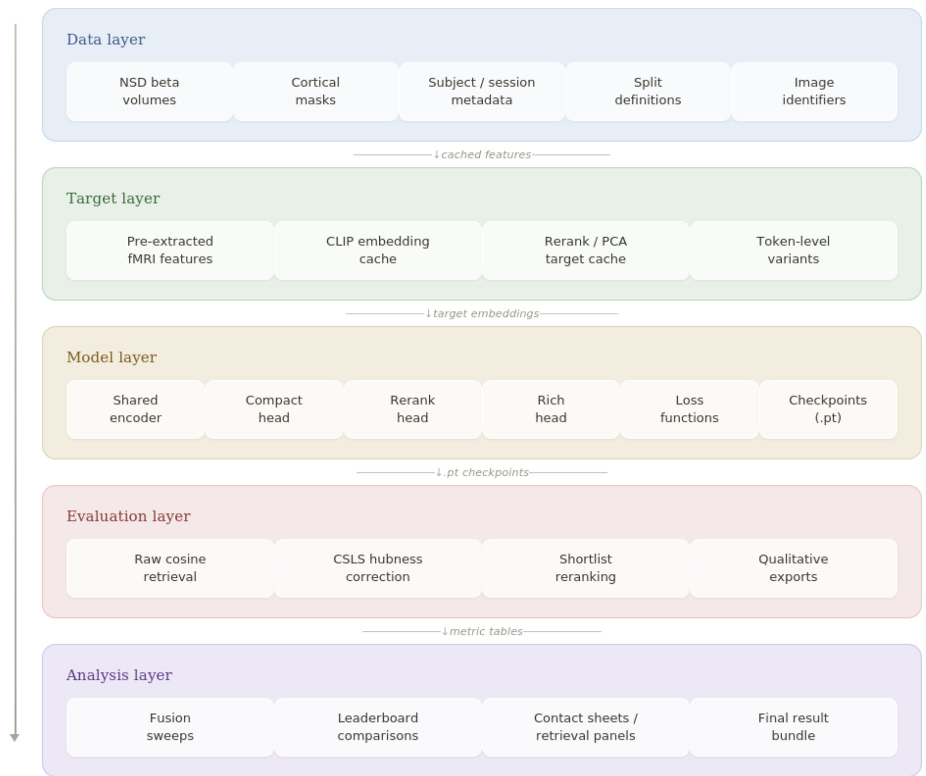


Figure 4.1: Software architecture of the research pipeline. The implementation is deliberately layered so that data preparation, target caching, training, evaluation, and qualitative export remain traceable and reproducible.

4.3 Configuration Management and Versioning

A recurring difficulty in research software is that experiments drift over time: hyperparameters change, split definitions evolve, and intermediate artifacts are overwritten. To avoid this, the pipeline is organized around versioned experiment configurations. Each major experiment wave is associated with an explicit configuration file that records data sources, target choices, architecture hyperparameters, loss weights, training settings, and evaluation options.

Versioning is especially important in this thesis because the experimental narrative depends on comparing multiple iterations rather than reporting only a single final run. When a new idea is tested, the goal is not merely to obtain a number but to understand whether a change in the score is caused by the underlying modelling idea or by a hidden change in the protocol. Configuration versioning therefore supports both reproducibility and interpretability.

4.4 Implementation of the Training and Evaluation Flow

The implementation flow can be summarized as follows. First, NSD-derived samples are loaded and normalized under a split-aware preprocessing regime. Second, target latents are either extracted on demand or retrieved from caches, reducing repeated compute and eliminating inconsistencies between experiments. Third, the selected decoder is trained with a loss appropriate to the experiment family. Fourth, validation-time retrieval is computed on a fixed gallery, together with CSLS correction and shortlist reranking when applicable. Finally, results are aggregated into human-readable summaries and compared against previous runs.

A useful abstraction is to think of the whole system as a deterministic map

$$\mathcal{P} : (\mathcal{D}, \mathcal{C}, \mathcal{A}) \longmapsto \mathcal{R}, \tag{4.1}$$

where $\mathcal{D}$ denotes the data and split definition, $\mathcal{C}$ the configuration, $\mathcal{A}$ the cached artifacts and checkpoints, and $\mathcal{R}$ the resulting metrics and qualitative outputs. The advantage of this view is that it clarifies what must remain fixed when two experiments are compared.

4.5 Validation and Testing Strategy

In the present project, testing took several complementary forms:

- **split-integrity checks**, ensuring that repeated images or derived artifacts cannot leak across train and validation sets;
- **shape and alignment checks**, verifying that every prediction matrix corresponds exactly to its intended ground-truth target matrix;
- **metric sanity checks**, confirming that retrieval metrics behave as expected on controlled inputs;
- **checkpoint and cache verification**, ensuring that the correct artifacts are loaded during evaluation;
- **qualitative inspection**, used to detect failure modes that aggregate metrics can hide.

This testing strategy is substantial enough to be regarded as part of the thesis contribution. Several improvements in the final system did not come from inventing a new loss function, but from identifying and eliminating hidden sources of inconsistency. In that sense, rigorous validation acted as a performance multiplier.

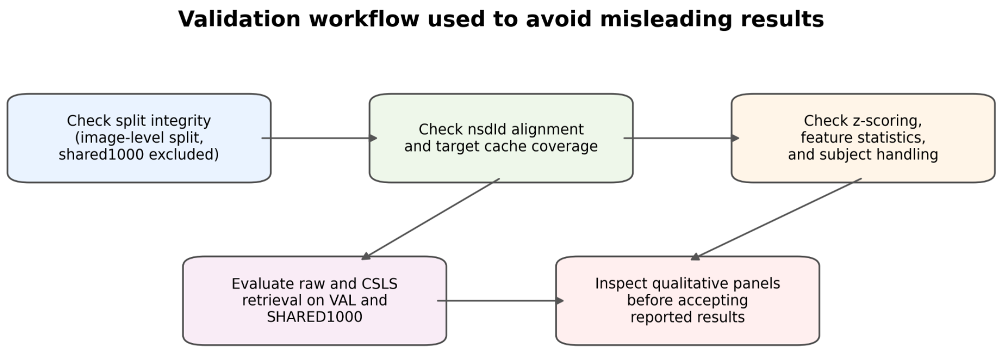


Figure 4.2: Validation workflow used throughout the thesis. Raw metric values are only trusted after split integrity, alignment, normalization, and qualitative inspection have been verified.

4.6 Reproducibility, Artifact Management, and Reporting

Reproducibility in this thesis is supported by explicit artifact management. Targets, checkpoints, predictions, and summary outputs are stored as named and versioned artifacts. This makes it possible to revisit a result months later and still recover its provenance. Such traceability is essential in a long-running experimental project, especially when results evolve across many versions.

The reporting layer is equally important. Tables, plots, retrieval examples, and diagnostic summaries are not treated as cosmetic additions but as part of the scientific record. For example, a metric table may reveal that rerank-only retrieval underperforms globally, while a shortlist-local diagnostic may show that the rerank head still contains useful discriminative information. Only by keeping both artifact provenance and reporting structure under control can such nuanced conclusions be justified.

Chapter 5

Experiments, Results, and Discussion

5.1 Experimental Strategy

The experimental strategy of the thesis is iterative. Instead of moving directly to the most complex architecture, the project first established strong baselines, then diagnosed failure modes, and only afterward introduced richer representation choices, multi-head decoders, and expert fusion. This approach is important because, in brain decoding, implementation errors and evaluation mismatches can easily masquerade as modelling limitations.

The history of the project reveals four broad phases. The first phase stabilizes the pipeline by validating sample–target alignment, correcting training pathologies, and confirming that evaluation measures true image-specific retrieval under a leakage-free protocol. The second phase investigates richer target spaces and multi-head retrieval design, culminating in the V30/V32 family where compact retrieval, reranking, and rich regression are explicitly separated. The third phase introduces expert complementarity: a stronger V35 base is combined at inference time with a complementary frozen legacy expert, providing the V35 plus N1v28a fixed-fusion milestone. The fourth phase extends this idea to a cross-architecture, cross-target-space score fusion that uses a 197K token-space model with MC-TTA and two complementary 768-D experts, yielding the final exported retrieval system reported in the thesis.

5.2 Evaluation Protocol

All experimental comparisons in the thesis are interpreted under a fixed retrieval-oriented evaluation protocol. The main endpoints are validation Recall@1 and SHARED1000 Recall@1, with CSLS-corrected retrieval used whenever hubness is relevant. This

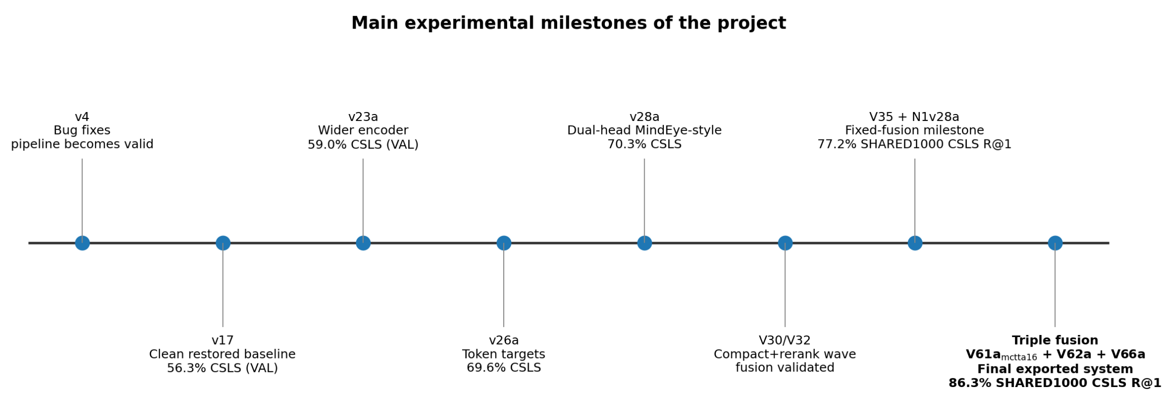


Figure 5.1: Main milestones of the experimental trajectory. The figure is not a strict apples-to-apples metric plot; rather, it highlights the turning points that changed the project: debugging the pipeline, widening the encoder, moving to token targets, validating dual-head decoding, establishing compact–rerank decomposition, freezing the V35 plus N1v28a fixed fusion as a methodological milestone, and finally extending it into the cross-architecture triple score fusion that becomes the exported retrieval system.

choice reflects the actual scientific question of the thesis: whether the predicted latent representation is discriminative enough to recover the correct stimulus image from a gallery.

The evaluation protocol is deliberately conservative. The validation split is used for model selection, fusion selection, and qualitative inspection; SHARED1000 is treated as the external benchmark split. When multiple experts are combined, the chosen recipe is selected on validation and then frozen before being applied to SHARED1000. This means that the final best system does not rely on test-time tuning.

5.3 Qualitative Interpretation of the Experimental Evolution

A major lesson of the project is that progress came more from disciplined diagnosis of bottlenecks than from arbitrary increases in architectural complexity. Early debugging stages exposed critical issues such as sample–image misalignment and unstable directional training. Later stages showed that richer target spaces improved the amount of recoverable information, but also created new geometric problems in retrieval. This led to a more modular formulation of the decoding task.

The V30/V32 wave provided the central architectural insight: a single representation should not be forced to solve global shortlist formation, local reranking, and generative supervision equally well. A compact space is beneficial for stable shortlist generation, a dedicated rerank space is useful for local correction among already plausible candidates, and a richer regression head preserves high-capacity information for downstream qualitative decoding. The V35 plus frozen legacy N1v28a fixed fusion built on this decomposition and established a strong milestone for inference-time expert combination. The final exported system preserves the same decomposition logic and extends it further by replacing two-expert legacy fusion with a frozen cross-architecture triple score fusion across different target spaces, which improves the SHARED1000 retrieval result without retraining the backbone.

Experimental wave	**Main question**	**Main lesson**
Baseline stabilization	Is the pipeline aligned and leakage-free?	Trustworthy experiments require correct sample/target alignment before any architecture scaling
Latent-space enrichment	Do richer or more structured targets help?	Target quality matters, but richer spaces also create harder retrieval geometry and stronger hubness effects
Multi-head retrieval design	Should one model solve all retrieval roles at once?	Separating compact retrieval, reranking, and rich regression is more effective than forcing one universal latent space
Compact–rerank fusion	Can reranking act as a corrective signal?	Yes; shortlist-local fusion improves over compact retrieval alone and validates the multi-stage design
Frozen expert fusion	Can complementary experts improve the exported system without retraining the whole backbone?	Yes; the V35 compact expert combined with the legacy N1v28a expert under fixed normalized-weighted fusion establishes a strong milestone
Cross-architecture triple fusion	Can experts from different target spaces and different architectures be combined at inference time?	Yes; a frozen triple score fusion across a 197K token-space model with MC-TTA and two complementary 768-D experts becomes the final exported system

Table 5.1: High-level summary of the major experimental waves in the thesis.

5.4 Final Retrieval Results

The final exported system is neither the early V30 compact–rerank fusion nor the V35 plus N1v28a fixed fusion taken alone, even though both waves remain methodologically central. It is a frozen score-level combination of three retrieval experts spanning different target spaces and inductive biases: a 197K token-space model (V61a) evaluated with MC-TTA-16, a 768-D CLS model (V62a), and a 768-D ROI-pretrained model (V66a). The weights $w_1/w_2/w_3 = 0.7/0.1/0.2$ and the CSLS neighbourhood size $k = 3$ are chosen on validation and frozen before SHARED1000, so the headline result does not depend on test-time tuning. The MC-TTA expert dominates each individual ranking; the two 768-D experts contribute as auxiliary score streams whose aggregate effect is consistent with complementary correction, although per-trial complementarity is not claimed because the corresponding fused ranks were not exported.

The headline retrieval endpoint of the thesis is CSLS Recall@1 on SHARED1000. Validation numbers were used during model selection and fusion weighting and are discussed where relevant, but they are not reported as headline results in Table 5.2: shared-image overlap between pretraining and validation splits in the V60–V66 lineage can inflate validation metrics relative to the strictly held-out benchmark, which is why the SHARED1000 result is treated as the paper-grade endpoint. Likewise, 197K token-space numbers and 768-D CLS numbers are not directly comparable on raw percentage points alone because the two spaces differ in difficulty and hubness geometry; CSLS partially mitigates this mismatch but does not remove it.

Table 5.2 separates four ideas that should not be conflated: the V30 compact–rerank wave, the V35 plus N1v28a fixed-fusion milestone, the stronger single-model 197K extension with MC-TTA, and the final frozen cross-architecture triple fusion.

Relative to the previous V35 plus N1v28a milestone the improvement on SHARED1000 is $+9.1$ percentage points; relative to V61a as a single non-TTA model it is $+7.2$ percentage points; and relative to V61a with MC-TTA-16 it is $+3.2$ percentage points. These gaps are large enough to motivate a complementarity interpretation: the aggregate metrics are consistent with the three experts contributing partially different information across target spaces and architecture families, which the frozen score-level fusion exploits without retraining the backbone.

Final cross-architecture fusion extension

Beyond the V30/V35 multi-head decomposition and the V35 plus N1v28a two-expert fixed fusion, the final extension explores whether retrieval can still be improved at inference time by combining experts that differ along two additional axes: target space dimensionality and architectural family. V61a is the strongest single model in the 197K token-space (ViT-L/14 grid) V60–V66 lineage; on top of V61a, MC-TTA-16 averages the predicted embeddings of 16 stochastic forward passes with dropout left active at inference and re-normalises the mean on the unit hyper-

System	Raw R@1	CSLS R@1	Interpretation
V30e compact head	—	49.2%	First strong compact shortlist head in the V30 wave
V30e compact + rerank fusion	—	51.5%	Validates rerank information inside a strong shortlist
V35 + N1v28a fixed fusion	—	77.2%	Previous best; milestone for two-expert fixed fusion
V61a (single 197K)	66.0%	79.1%	Strongest single non-TTA model
V64a (continued V61a)	70.6%	78.6%	Slightly below V61a despite continued training
V61a + MC-TTA-16	67.3%	83.1%	Strongest single-model variant
V62a (768-D CLS)	39.9%	47.5%	Weak alone; auxiliary expert in the frozen fusion
V66a (768-D ROI pretrain)	35.5%	39.1%	Weak alone; auxiliary expert in the frozen fusion
Triple fusion V61a$_{\text{mctta16}}$ + V62a + V66a	**72.1%**	**86.3%**	**Final exported system; R@5 = 98.0%, MRR = 0.914, median rank 1**

Table 5.2: Final retrieval results on SHARED1000, reported as Recall@1 with and without CSLS hubness correction. The V30/V32 compact–rerank wave and the V35 plus N1v28a fixed fusion are retained as methodological milestones, while the final exported system is the frozen cross-architecture triple score fusion with weights $0.7/0.1/0.2$ and CSLS neighbourhood size $k = 3$. SHARED1000 CSLS Recall@1 is treated as the paper-grade endpoint.

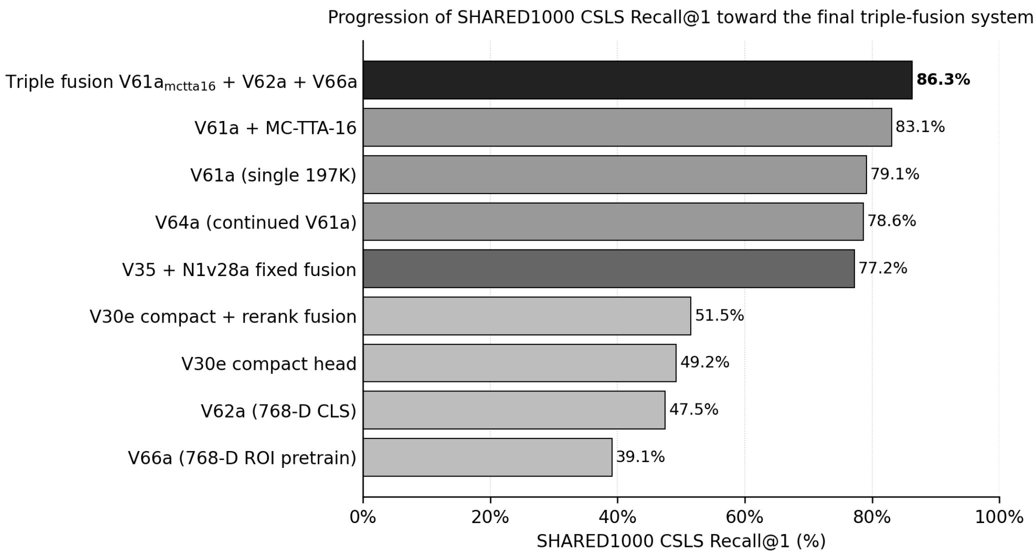


Figure 5.2: Progression of SHARED1000 CSLS Recall@1 toward the final frozen triple-fusion system. The V35 plus N1v28a fixed fusion is shown as a milestone; the final exported triple fusion of V61a with MC-TTA-16, V62a, and V66a is the highest bar.

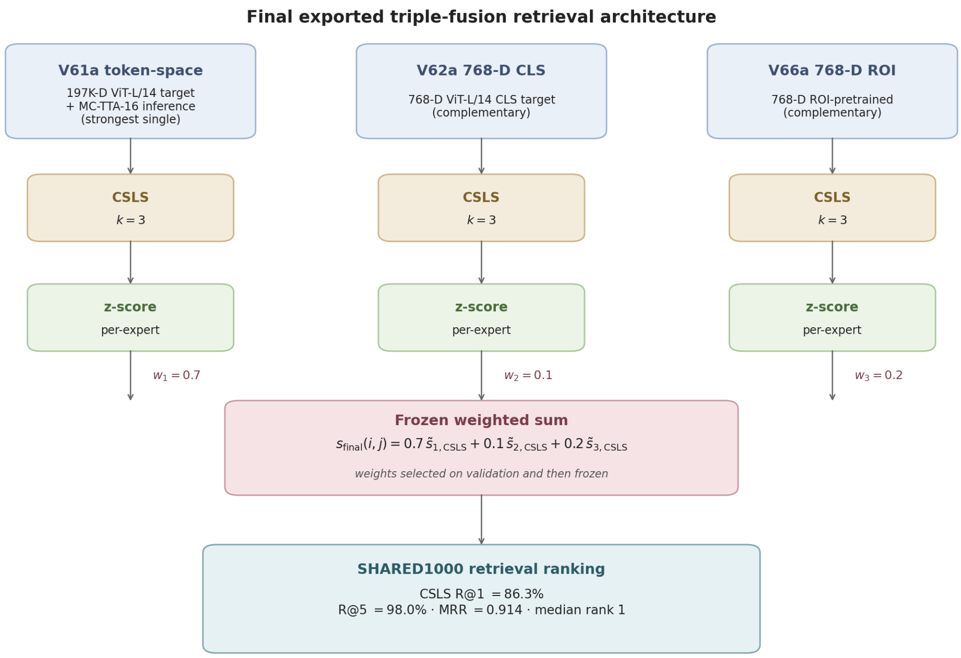


Figure 5.3: Final exported triple-fusion retrieval architecture. The three experts (V61a token-space with MC-TTA-16, V62a 768-D CLS, V66a 768-D ROI) each produce a CSLS-corrected retrieval score at neighbourhood size $k = 3$, the scores are z-score normalized per expert, and a frozen weighted sum with $w_1/w_2/w_3 = 0.7/0.1/0.2$ yields the SHARED1000 retrieval ranking. The fusion is score-level and inference-time only; it does not produce a unified fused latent embedding.

sphere, which reduces per-trial prediction variance at $16\times$ inference-time compute. V62a (768-D CLS) and V66a (768-D ROI-pretrained) are comparatively weak alone but operate in geometrically distinct spaces and bring complementary inductive biases. The three experts are combined under frozen z-score-normalised CSLS scores with the weights and neighbourhood size given in Equation (3.11); the combination is a score-level inference-time fusion and does not produce a unified fused latent embedding, so retrieval-geometry claims for the fused system are deliberately deferred to the V62a single-expert analyses of Section 5.5. Figure 5.3 summarises the three-expert pipeline.

Why score-level triple fusion improves retrieval

The retrieval gain in Table 5.2 should not be read as the additive effect of stacking similar experts: the three components of the final system differ in target dimensionality, training objective, and inductive bias, so they fail in measurably different ways at the aggregate level. Table 5.3 reports the per-expert SHARED1000 numbers. The 197K token-space expert V61a with MC-TTA-16 dominates each individual ranker, yet the frozen combination still exceeds it on the held-out benchmark even though the two weaker 768-D experts retain non-negligible weights $w_2 = 0.1$ and $w_3 = 0.2$ after validation tuning. That pattern is consistent with complementary correction at the system level. For the bulk of SHARED1000 we do not export per-trial fused ranks in this thesis — the four illustrated trials in Figures 5.5 and 5.6 are the only exception — and we therefore make no per-trial claim about which specific V61a errors are corrected by V62a or V66a across the full benchmark.

Expert	CSLS R@1	CSLS R@5	CSLS MRR	Median rank	Role in fusion
V61a, 197K token-space, MC-TTA-16	83.1%	96.9%	0.892	1	Dominant ranker ($w_1 = 0.7$)
V62a, 768-D CLS	47.5%	80.3%	0.621	2	Auxiliary expert with $w_2 = 0.1$
V66a, 768-D ROI-pretrained	39.1%	73.8%	0.545	2	Auxiliary expert with $w_3 = 0.2$
Triple fusion (frozen)	$\mathbf{86.3}\%$	$\mathbf{98.0}\%$	$\mathbf{0.914}$	**1**	Headline exported system

Table 5.3: Per-expert SHARED1000 retrieval metrics under the same 1000-image target-embeddings gallery as the headline endpoint, and the frozen-weight triple-fusion result they produce. Numbers are reproduced verbatim from the exported fusion summary; the gain over the strongest single expert is $+3.2$ percentage points on CSLS Recall@1 and $+1.1$ percentage points on CSLS Recall@5.

Three architectural decisions are responsible for converting raw per-expert scores into a usable joint ranking. First, the per-expert CSLS correction at neighbourhood size $k = 3$ removes the hubness asymmetries that would otherwise let a small set of generic images dominate retrievals in each space (Section 5.5 quantifies this effect for the analyzed expert). Second, the per-expert z-score normalization across the SHARED1000 gallery places all three score streams on a comparable, dimensionless scale before they are summed; raw CSLS scores live in different ranges in the 197K token-space and in the 768-D CLS space, and combining them without normalization would let scale, not information content, decide the result. Third, the weights $w_1/w_2/w_3 = 0.7/0.1/0.2$ are selected on the validation split by sweeping a small simplex, then frozen before the SHARED1000 evaluation; they are not tuned on the held-out benchmark. The weight on V61a reflects its much stronger standalone retrieval, while the smaller weights on V62a and V66a allow auxiliary score streams to influence the final ranking without overwhelming the dominant expert.

The aggregate gain is most visible in the near-miss regime: CSLS Recall@5 rises from 96.9% for the strongest single expert to 98.0% for the fusion, mean reciprocal rank rises in parallel, and the median rank stays at 1. This signature is more consistent with complementary correction than with redundant ensembling, because redundant experts would mainly reinforce the same top-1 mistakes rather than shorten the tail of slow-decaying ranks. Two caveats apply: the gain holds under a fixed gallery and a fixed weight choice and is not a guarantee about other settings, and because the fusion operates on scalar scores rather than on a unified embedding, the geometric and manifold-level analyses of Section 5.5 apply to the V62a single expert, not to the fused system.

5.5 Generalization, Robustness, and Manifold Alignment

The headline retrieval gain reported in Table 5.2 answers a narrow but well-defined generalization question: given an fMRI response to a held-out stimulus image, can the decoder rank that image first in a held-out gallery? This subsection records the design choices behind that question and reports auxiliary geometric evidence for the broader claim that the learned fMRI-to-CLIP mapping preserves semantic structure, rather than merely memorizing top-1 identities.

What counts as generalization evidence. SHARED1000 is treated as the main paper-grade endpoint because it is held out from training and validation under the evaluated subject-specific SHARED1000 protocol, and because the chosen fusion recipe is frozen on validation before being applied. Splits are stimulus-level: each unique stimulus image belongs to exactly one of train, validation, and SHARED1000, so repeated fMRI trials of the same image cannot leak across the three sets. Validation Recall@1 is used for model selection and for choosing the triple-fusion weights and CSLS neighbourhood size, but it is not reported as the headline endpoint: shared-image overlap with earlier pretraining phases in the V60–V66 lineage can inflate validation retrieval relative to the strictly held-out benchmark, as already noted in Section 5. CSLS is preferred over raw cosine because high-dimensional CLIP retrieval is known to suffer from hubness, and the manifold analyses below quantify the size of this effect on the analyzed models.

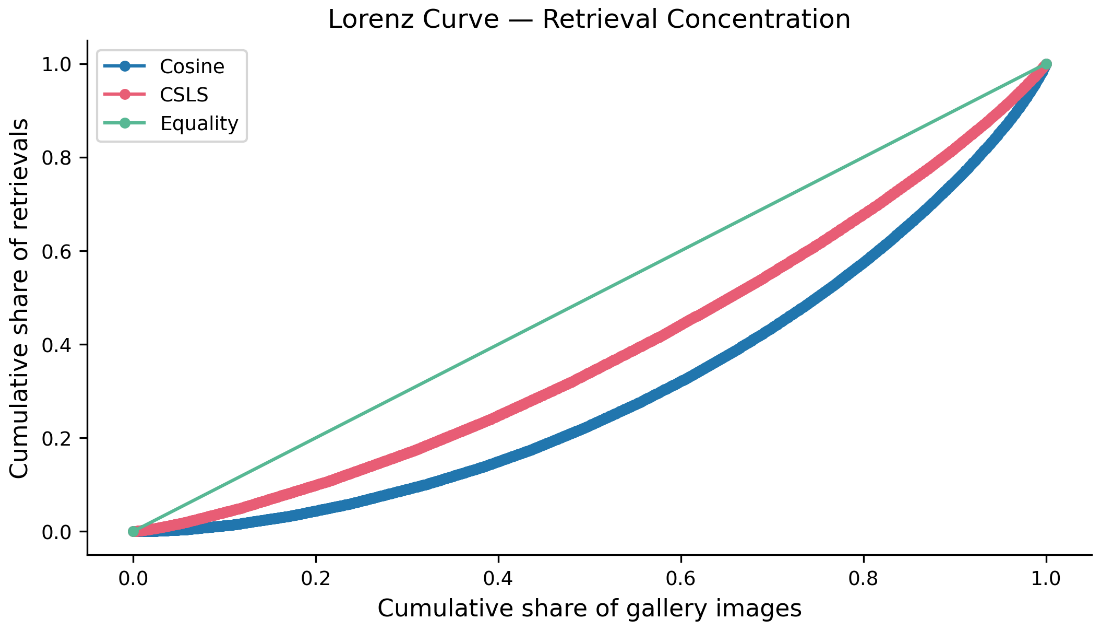
Scope of the manifold-alignment analyses. The geometric analyses summarized in Table 5.4 were computed on the V62a single model under the official SHARED1000 target-embeddings protocol with a 1000-image gallery, and supplementarily on the V62a validation split with a 10000-image gallery. They therefore characterize the 768-D CLIP-CLS retrieval geometry of one expert that participates in the final triple fusion; they do not directly characterize the fused system, because the triple fusion is performed at score level and does not produce a unified fused 768-D embedding.

The analyses are presented as auxiliary evidence that the learned fMRI-to-CLIP mapping preserves semantic structure, not as a substitute for the SHARED1000 retrieval endpoint.

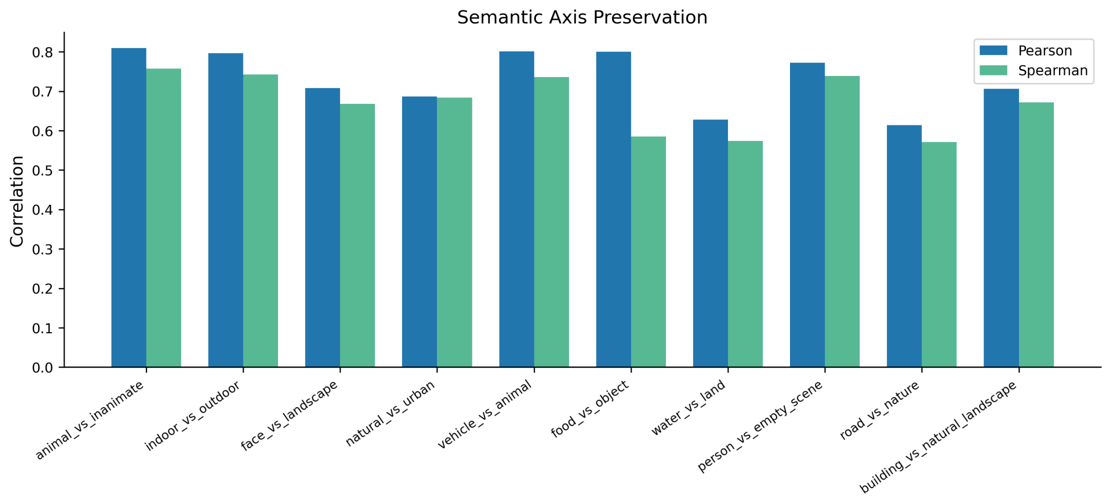
Evidence type	Metric (source)	What it supports	Caveat
Held-out retrieval	Triple-fusion SHARED1000 CSLS R@1 $= 86.3\%$ (Table 5.2)	Stimulus-level generalization on the headline endpoint	The headline endpoint; the only direct claim about the final exported system
Hubness control	V62a SHARED1000 hubness Gini cosine $0.386 \to$ CSLS 0.226	CSLS substantially reduces hubness in the analyzed 768-D space	Auxiliary analysis on the V62a expert only
Representational geometry	V62a SHARED1000 RSA Spearman $\rho = 0.389$, bootstrap CI $[0.387, 0.392]$ (val. $\rho = 0.583$)	Predicted and target RDMs share a stable monotonic structure on the harder benchmark	Single-expert evidence; magnitude drops on the harder split as expected
Local neighborhood preservation	V62a SHARED1000 Overlap@10 $= 0.40$ vs random 0.01, lift $\approx 40\times$ (val. lift $\approx 514\times$)	Local CLIP-space neighborhoods are preserved well above chance	Lift depends on gallery size; the SHARED1000 lift is the conservative number
Language-space interpretability	V62a SHARED1000 CLIP text-probe agreement@5 $= 0.833$ (29 concepts) and mean semantic-axis Spearman $= 0.673$ across 10 axes	Decoded embeddings agree with CLIP language concepts and preserve interpretable concept directions	Axis quality depends on concept choice; not literal thought decoding

Table 5.4: Generalization and manifold-alignment evidence. The retrieval row is the headline endpoint for the final exported triple-fusion system; the remaining rows are auxiliary geometric analyses computed on the V62a single model under the SHARED1000 target-embeddings protocol (1000-image gallery), with validation values added in parentheses for context.

What this evidence does and does not say. Taken together, the rows of Table 5.4 are consistent with a decoder that learns a structured fMRI-to-CLIP mapping rather than one that memorises a small set of training images: hubness is substantially reduced under CSLS, local neighbourhoods are preserved well above chance, the representational dissimilarity structure remains positively rank-correlated with the target, and decoded embeddings agree with CLIP text probes in the large majority of SHARED1000 trials. At the same time, several pieces of evidence are deliberately omitted because they are weak or unavailable in the current artifacts: standalone confidence based on the von Mises–Fisher concentration parameter κ is close to chance on validation and is not exported on SHARED1000; manifold-density-only separation between correct and incorrect predictions is rejected in the V62a claim



(a) Hubness Lorenz curves. The CSLS curve is closer to the equality line than the cosine curve, illustrating the reduction of the V62a SHARED1000 hubness Gini from 0.386 to 0.226.



(b) Per-axis Pearson and Spearman correlations of decoded-vs-target projections for the ten interpretable CLIP semantic axes. *Indoor vs. outdoor* is the best-preserved axis; *water vs. land* is the weakest.

Figure 5.4: Two auxiliary visual probes of the V62a single-expert SHARED1000 geometry (1000-image gallery), corresponding to two rows of Table 5.4. Both panels characterize the V62a single expert and not the fused system, in line with the scope statement of this section.

tests; and reliability-aware selective decoding works on validation but does not transfer to SHARED1000. We therefore do not claim selective prediction or κ-based uncertainty as part of the generalisation story.

Robustness checks supported by the current artifacts. Three lightweight robustness checks are consistent with the headline numbers. First, the RSA Spearman estimate on V62a SHARED1000 has a narrow bootstrap confidence interval (width ≈ 0.005), which suggests that the geometric signal is stable across resamplings rather than driven by a small set of outlier trials. Second, the triple-fusion system reaches R@5 $= 98.0\%$ and median rank 1 on SHARED1000 (Table 5.2), so the correct stimulus remains within the top five for 98.0% of SHARED1000 trials; equivalently, most remaining non-top-1 errors are still near-misses rather than complete retrieval failures, which is the gallery-size-related robustness signal most directly testable from the available outputs. Third, CSLS provides the same qualitative hubness improvement across all three analyzed retrieval spaces (V61a token space, V62a 768-D CLS, V66a 768-D ROI), which suggests that the hubness correction is a property of the retrieval protocol rather than of any single expert. We do not perform gallery-size sweeps beyond what these artifacts already report.

Beyond top-1: semantic axes, interpolation, and counterfactual robustness

The retrieval and neighborhood metrics summarized in Table 5.4 characterize the V62a expert's behaviour at a single point along the CLIP manifold (the predicted embedding for one trial). Three additional analyses, all computed on the same V62a SHARED1000 protocol, probe whether the surrounding latent geometry is also semantically organized rather than only locally correct at the nearest neighbour. Each analysis uses interpretable CLIP-space constructs and does not claim to read out neural representations themselves; the constructs are operational objects in image-text embedding space, not literal cognitive categories.

Language-space interpretation of decoded embeddings. Before reporting the three aggregate analyses, Table 5.5 shows three representative SHARED1000 cases drawn verbatim from the exported semantic-probe artifact. Each row lists the top-three CLIP text concepts ranked by similarity to the V62a predicted embedding and the top-three concepts ranked by similarity to the corresponding ground-truth CLIP image embedding, evaluated against the same 29-concept probe library. These per-trial probes are not retrieval outputs; they are a language-space view of where each decoded embedding lies, used here as an auxiliary interpretability diagnostic for the aggregate agreement@5 statistic of 83.3%.

Sample	Top-3 decoded probes	Top-3 target-image probes	Interpretation
1	kitchen; indoor room; food	kitchen; food; indoor room	Top-3 concept sets coincide; decoded top-1 matches target top-1.
7	sports; forest; natural scene	sports; human body; person	Top-1 matches; secondary concepts emphasize scene context (decoded) vs. actor context (target).
8	dog; animal; cat	cat; car; animal	Animal category is preserved; instance-level identity is confused (decoded top-1 is the wrong species but the target top-1 *cat* appears in the decoded top-3).

Table 5.5: Three representative SHARED1000 trials drawn from the V62a semantic-probe export, illustrating the range of behaviours summarized by the aggregate agreement@5 = 0.833 statistic. Sample indices match the exported `semantic_probe_examples` artifact; values are reproduced verbatim. These examples are language-space probes of decoded V62a embeddings, not retrieval outputs of the triple-fusion system.

Semantic axes. Ten interpretable directions are constructed in CLIP text space as differences of paired prompts, taking the form

$$\mathbf{a}_{p|q} = \mathbf{t}_p - \mathbf{t}_q, \tag{5.1}$$

where $\mathbf{t}_p$ and $\mathbf{t}_q$ are CLIP text embeddings of paired concepts such as *indoor / outdoor* or *animal / inanimate*. Ten such axes are constructed, covering scene context, semantic category, and salient object properties. For each held-out trial the V62a predicted embedding is projected onto every axis and compared with the projection of the corresponding ground-truth CLIP image embedding. Across the ten axes the mean Pearson correlation between predicted and target projections is 0.732 and the mean Spearman correlation is 0.673; *indoor vs. outdoor* is the best-preserved axis and *water vs. land* is the weakest, as illustrated in Figure 5.4b. This pattern is consistent with the cortical-organisation expectation that scene-level distinctions are decoded more reliably than fine compositional distinctions; the thesis does not separate region-of-interest contributions to specific axes and therefore stops short of region-specific claims.

Geodesic interpolation on the unit hypersphere. Twenty pairs of held-out V62a embeddings are interpolated by normalized spherical interpolation,

$$\mathbf{z}(t) = \mathrm{slerp}(\mathbf{z}_A, \mathbf{z}_B; t), \qquad t \in [0, 1], \tag{5.2}$$

in 21 equally spaced steps. At each step the closest CLIP text concepts and the nearest-neighbour gallery images are recorded. The mean smoothness of the resulting paths, measured as one minus the normalized step-to-step displacement, is 0.9998. Across the 20×21 trajectories the total number of top-concept transitions is 26 — that is, 26 step-intervals on which the closest CLIP text concept changed at all — and these transitions are distributed gradually along the paths rather than concentrated at a few discontinuities. The stricter *abrupt-transition rate*, defined as the fraction of step-intervals on which the top-1 concept changes with a single-step displacement above the smoothness threshold, is only 0.75%, so the large majority of the 26 concept transitions are smooth rather than discontinuous. These numbers are consistent with the analyzed CLIP-space region behaving as a continuous semantic surface rather than as a collection of disconnected cluster centroids; they characterize the local geometry around held-out embeddings only and are not a global manifold-topology guarantee.

Counterfactual semantic editing. The same ten semantic axes are used to construct counterfactual variants of each predicted embedding: $\mathbf{z}^{\mathrm{cf}}_{a,\alpha} = \mathrm{normalize}(\mathbf{z} + \alpha\, \mathbf{a})$ for axis $\mathbf{a}$ and scalar magnitude α. For each of 24 held-out trials and each of the 10 axes, the smallest magnitude that flips the dominant text-probe concept is recorded; in every case a finite magnitude exists, with a mean robustness margin of about 0.5. The interpretation is that decoded embeddings sit in regions of CLIP space where small directed perturbations along language-defined axes shift the dominant semantic label in a controlled way, rather than collapsing into a generic high-density attractor. The robustness margin summarises how far the decoded embedding is from the nearest semantic boundary and is used descriptively, not as a calibrated confidence score.

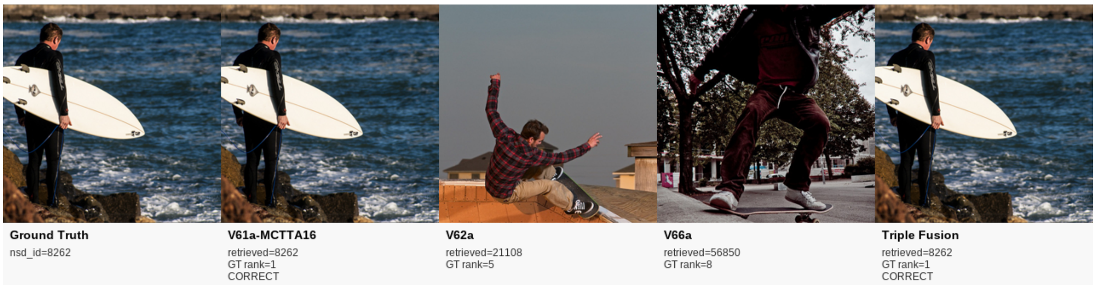
Aggregate composite score. The V62a SHARED1000 manifold report computes a harmonic-mean-style composite of retrieval, neighborhood overlap, RSA, and on-manifold density. Its value, $\mathrm{NMAS} = 0.412$, is reported here only as a single-number summary of the analyses above for cross-reference with the manifold report; it is not used as a primary thesis metric, and it does not replace the SHARED1000 CSLS Recall@1 endpoint. The reason is that aggregate manifold scores conflate evidence types that the thesis prefers to keep distinct: retrieval performance, geometry preservation, and uncertainty signals each fail in different ways, and a single composite obscures which subcomponent is responsible when the composite changes.

What these analyses add to the claim register. The three new analyses are mutually consistent and extend the existing evidence in a specific way: where neighborhood overlap and RSA quantify how well the decoder preserves the order and identity of nearest neighbours, the axis, interpolation, and counterfactual analyses quantify how well it preserves *directions* and *transitions* in the same latent space. They support the same broader claim that the decoded CLIP-space mapping is semantically structured rather than memorized, and they share the same scope: they are properties of the V62a single expert and do not generalize automatically to the fused system, because the fusion is score-level. We avoid stronger claims (perfect manifold reconstruction, region-specific axis attribution, generative-quality semantic editing) and present these analyses strictly as auxiliary geometric evidence consistent with the headline retrieval result.

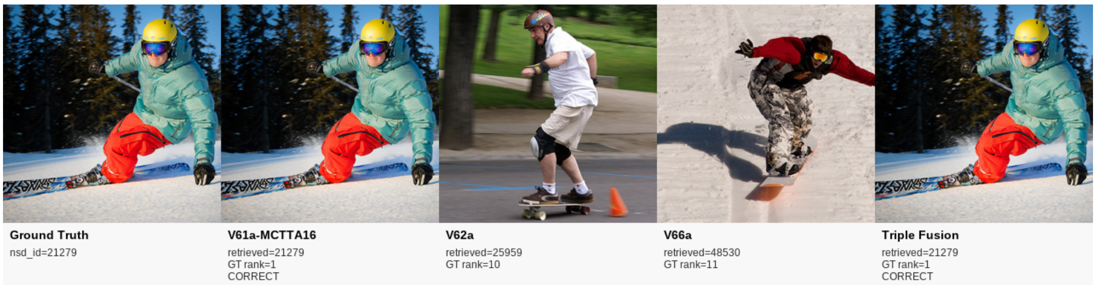
5.6 Qualitative Retrieval Evidence

The qualitative retrieval panels below show direct triple-fusion top-1 outputs alongside the individual experts, for four representative SHARED1000 stimuli. Each panel displays five columns: the ground-truth stimulus, the V61a-MCTTA16 expert's top-1 retrieval (197 K-D token space, 16-draw MC-TTA), the V62a expert's top-1 (768-D CLS space), the V66a expert's top-1 (768-D ROI pretrain CLS), and the triple-fusion top-1 after z-scored score-level fusion with CSLS (k=3, weights $0.7/0.1/0.2$).

Across all four examples, V61a-MCTTA16 — the dominant expert at 83.1% CSLS Recall@1 — retrieves the correct stimulus at rank 1 on its own. In three of the four cases the weaker experts (V62a at 47.5%, V66a at 39.1%) miss with semantically plausible but incorrect alternatives; on the fruit-bowl example V66a also retrieves the correct image at rank 1 while V62a misses only narrowly at rank 2. In every panel the triple fusion preserves the rank-1 retrieval, illustrating that the frozen score-level combination does not perturb V61a's already-correct results when the weaker experts disagree. These four panels are therefore representative of the 83.1% of SHARED1000 trials in which V61a alone is already correct; the aggregate gain from 83.1% to 86.3% must instead arise on the disjoint subset of trials where V61a's top-1 is wrong, which is not visualised here because per-trial fused ranks across the full benchmark were not exported.



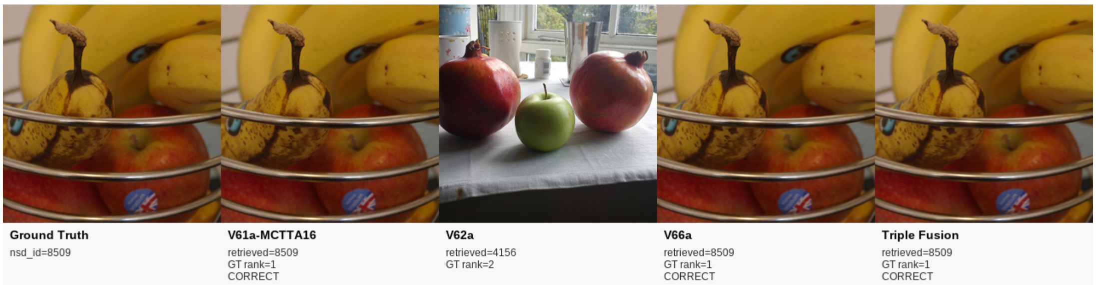
(a) Surfer query (NSD 8262). V61a-MCTTA16 retrieves the correct image (rank 1). V62a returns a different scene (rank 5); V66a returns another incorrect image (rank 8). The triple fusion recovers the ground truth at rank 1.



(b) Skier query (NSD 21279). V61a-MCTTA16 retrieves the correct image (rank 1). V62a misses (rank 10); V66a misses (rank 11). The triple fusion recovers the ground truth at rank 1.

Figure 5.5: Qualitative retrieval successes (I). Each panel shows, from left to right: ground-truth stimulus, V61a-MCTTA16 top-1, V62a top-1, V66a top-1, and Triple Fusion top-1. The weaker 768-D experts (V62a, V66a) fail with semantically plausible but incorrect alternatives, while V61a-MCTTA16 and the triple fusion both recover the exact image at rank 1.

(a) Seagull query (NSD 55649). V61a-MCTTA16 retrieves the correct image (rank 1). V62a misses (rank 4); V66a misses (rank 12). The triple fusion recovers the ground truth at rank 1.



(b) Fruit-bowl query (NSD 8509). V61a-MCTTA16 and V66a both retrieve the correct image (rank 1). V62a narrowly misses (rank 2). The triple fusion recovers the ground truth at rank 1. This convergence among experts is characteristic of the easy regime.

Figure 5.6: Qualitative retrieval successes (II). Column order as in Figure 5.5: GT, V61a-MCTTA16, V62a, V66a, Triple Fusion. The seagull example extends the success pattern; the fruit-bowl example shows expert convergence where even V66a retrieves the correct image.

5.7 Reconstruction Add-on: What the Qualitative Evidence Actually Supports

The thesis treats reconstruction more conservatively than retrieval. The retrieval system is the primary quantitative contribution, whereas the diffusion add-on is a small-scale extension that uses the final triple-fusion top-1 retrievals as the img2img anchor for each of the 16 selected SHARED1000 examples. The improved pipeline conditions SD 2.1 on the fMRI-predicted CLIP embedding (replacing the empty prompt used in the baseline), generates 16 candidates per example with uncertainty-aware strength derived from a per-example confidence score, and selects the best candidate by CLIP cosine similarity to the predicted embedding. The evidence supplied by the selected examples and the 16-example evaluation summary supports three conclusions.

First, *exact retrievals should not be harmed*. In the perfect bucket, the final selected output remains identical to the anchor image. This is the correct behaviour: when the retrieval stage is already exact, the generative module should preserve it rather than rewrite it.

Second, *the effect of diffusion on non-perfect cases is bucket-dependent rather than uniformly positive*. On the good and hard buckets, diffusion improves pixel correlation and PSNR while regressing CLIP image similarity; on the near-good bucket the per-metric outcome is the most adverse of the three, with the largest CLIP image-similarity regression and a small pixel-correlation regression. The benefit is therefore neither universal across buckets nor uniform across metrics, and the reconstruction stage does not act as a reliable improvement over the retrieval anchor.

Third, *CLIP-conditioned best-of-N diffusion trades off low-level fidelity for semantic steering*. The improved diffusion pipeline injects the fMRI-predicted CLIP embedding as conditioning for SD 2.1 img2img, generates $N{=}16$ candidates per example, and selects the candidate with highest CLIP cosine similarity to the predicted embedding. Strength, guidance scale, and number of steps are derived per example from a confidence score $c \in [0, 1]$ defined as the margin between the CSLS top-1 and CSLS top-2 fused scores for the query, linearly mapped to the strength interval $[0, 0.30]$ (high confidence $\Rightarrow$ low strength, low confidence $\Rightarrow$ higher strength), with the guidance scale and number of denoising steps interpolated jointly on the same axis, rather than bucket-fixed routes. The full per-metric outcome is reported in Table 5.6: PSNR and AlexNet(2) gain in the mean, SSIM, LPIPS, CLIP image similarity, and AlexNet(5) regress, and pixel correlation is essentially flat. The pattern is interpretable: the CLIP-conditioned pipeline actively steers generation toward the predicted semantics, producing images that are more structurally diverse from the anchor but not uniformly closer to ground truth. The strongest per-example deviations are concentrated in the hard bucket, where the retrieval anchor itself is least similar to the ground truth.

The bucketed analysis refines the picture. Perfect cases remain perfect, validating the anchor-first controller. For the good bucket, diffusion improves pixel correlation ($+0.003$), PSNR ($+0.212$ dB), and AlexNet(2) ($+0.015$), while CLIP image similarity decreases (-0.013). For the hard bucket, PSNR improves most strongly ($+0.151$ dB), pixel correlation increases ($+0.007$), and AlexNet(2) gains $+0.017$, while CLIP similarity decreases (-0.033). The near-good bucket is the most adverse of the three: it shows the largest CLIP image-similarity regression (-0.042), a small pixel-correlation regression (-0.007), and an SSIM regression (-0.022), with only modest PSNR ($+0.051$ dB) and AlexNet(2) ($+0.013$) gains. This signature places the CLIP-conditioned pipeline on a different operating point along the fidelity-versus-exploration trade-off rather than on a uniformly stronger one: the predicted brain signal now actively steers generation, which is methodologically more meaningful than running diffusion against an empty prompt, but the per-metric outcome on this curated 16-example subset is mixed and the reconstruction stage still does not match the retrieval stage's contribution.

5.8 Ablation-Oriented Discussion

Although the revised thesis now reports the frozen cross-architecture triple fusion as the final exported retrieval system, with the V35 plus N1v28a fixed fusion retained as a methodological milestone, the negative results remain almost as informative as the positive ones. Several experimentally attractive ideas did not produce the expected gains:

- purely probabilistic directional formulations proved harder to stabilize than initially expected;
- richer latent targets improved pointwise expressivity but sometimes degraded retrieval geometry;
- anatomically structured or tokenized formulations were not automatically superior to strong full-vector baselines;
- learned global gates and residual rerankers easily overfit if expert disagreement patterns are not matched across train and evaluation conditions;
- CLIP-conditioned generative refinement with best-of-N selection improves low-level reconstruction metrics (PixCorr, PSNR, AlexNet-2) but at the cost of seman-

Metric (16-example subset)	Retr. mean	Diff. mean	Δ mean	Δ median	Interpretation
Pixel correlation	0.315	0.316	+0.001	−0.011	Mean is essentially flat; median is slightly negative
SSIM	0.358	0.350	−0.009	−0.010	Structural similarity decreases slightly
PSNR (dB)	9.382	9.520	+0.138	+0.046	Mean is pulled up by a small number of hard-bucket examples; median improvement is much smaller
LPIPS ↓	0.546	0.558	+0.012	+0.044	Perceptual distance increases; the median regression is larger than the mean
CLIP image similarity	0.747	0.725	−0.022	−0.016	Semantic steering shifts generation away from the anchor
AlexNet(2)	0.560	0.571	+0.011	+0.016	Early-layer perceptual features improve
AlexNet(5)	0.504	0.494	−0.010	−0.006	Late-layer perceptual features decrease slightly

Table 5.6: Retrieval-versus-diffusion summary for the 16 selected SHARED1000 examples using the improved CLIP-conditioned best-of-16 pipeline with uncertainty-aware strength. PSNR is computed over the 12 non-perfect examples (perfect cases yield infinite PSNR). Both mean and median deltas are reported because for $n = 12$ non-perfect examples the mean is sensitive to outliers; the comparison between the two columns indicates that the apparent PSNR gain is concentrated on a small number of examples. Numbers trace verbatim to `experimental_results/reconstruction_triple/metrics_summary.json`.

tic fidelity to the anchor, and the reconstruction stage still does not match the retrieval stage's contribution.

This ablation-oriented perspective is valuable because it prevents over-interpretation. Instead of claiming that every added modelling idea is useful, the thesis distinguishes between improvements that genuinely change the retrieval outcome and changes that are aesthetically appealing but practically weak under the current data regime.

5.9 Comparison to the Literature

The results of the thesis should be interpreted in context rather than in isolation. Recent state-of-the-art systems show that much larger gains are possible when one combines stronger shared-subject scaling, richer token targets, and more ambitious alignment strategies [13]. The present thesis does not yet match the strongest large-scale shared-subject systems, but it does answer a narrower and still meaningful question: within a retrieval-focused decoding setting, how should the available information be decomposed and fused?

In that narrower setting, the thesis provides three clear answers. First, compact retrieval, shortlist-local reranking, and rich regression should not be forced into one universal latent space; they are better treated as distinct roles. Second, once a strong compact expert exists, complementary expert fusion can still deliver large gains without retraining the whole system, as the V35 plus N1v28a milestone demonstrates. Third, additional gains remain available by extending the same frozen-fusion idea across different target spaces and architecture families and by combining it with single-model strengthening through MC-TTA. The final exported result of 86.3% CSLS Recall@1 on SHARED1000 is therefore not just a higher number; it is evidence that retrieval quality in this problem depends as much on how information is combined at inference time as on how it is first predicted. The system does not match the strongest large-scale shared-subject methods in the literature, and the comparison to those numbers must be made carefully because their galleries, splits, and CLIP backbones often differ.

5.10 Limitations

Despite the encouraging results, several limitations remain. The study is still constrained by subject scale, by the difficulty of cross-subject alignment, and by the fact that retrieval is quantitatively emphasized more strongly than reconstruction fidelity. The final best system also relies on a frozen inference-time score fusion rather than on a fully end-to-end fusion-aware training objective, and does not produce a unified fused latent embedding. The use of 16-draw MC-TTA on the dominant 197K expert increases inference cost relative to a single forward pass. Validation Recall@1 in the V60–V66 lineage can be inflated by shared-image overlap with pretraining splits, which is why SHARED1000 CSLS Recall@1 is treated as the paper-grade endpoint throughout this chapter. Finally, the diffusion add-on is evaluated only on a curated 16-example qualitative subset, so its conclusions should be interpreted as design evidence rather than as a definitive large-scale benchmark of the system's downstream generative behaviour.

These limitations do not weaken the current conclusions; rather, they clarify the next research steps. The thesis has reached a point where the main challenge is no longer basic pipeline validity, but how to preserve complementary expert signals while scaling toward stronger shared-subject and reconstruction-aware systems.

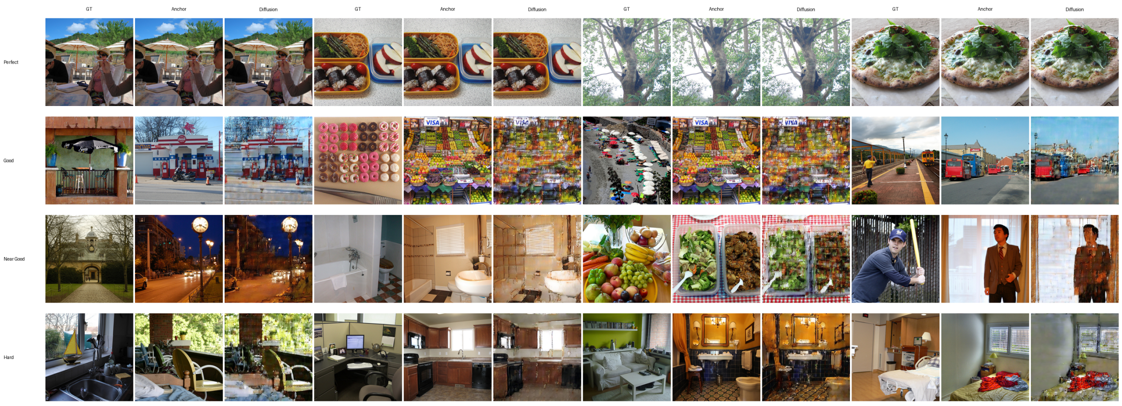


Figure 5.7: Representative CLIP-conditioned reconstruction examples anchored on the final triple-fusion top-1 retrievals. Perfect cases are preserved unchanged by the identity-pass route. Good, near-good, and hard cases use fMRI-predicted CLIP conditioning with best-of-16 candidate selection and uncertainty-aware strength. The diffusion outputs are visibly steered toward the predicted semantics rather than toward the anchor, consistent with the mean PSNR and AlexNet(2) gains and the corresponding CLIP image-similarity, SSIM, and LPIPS regressions reported in Table 5.6.

Chapter 6

Conclusions and Future Work

This thesis presented a complete study of neural decoding of visual perception from fMRI using CLIP-guided latent targets and retrieval-oriented modelling. The work treated the problem not simply as a regression task, but as a structured mapping from cortical activity to a visual representation space that can support both image retrieval and downstream generation. Across the project, the emphasis was placed equally on model design and on scientific debugging, because the validity of any claimed result depends on the reliability of the preprocessing and evaluation pipeline.

The final outcome is a strong retrieval-oriented system that integrates three levels of insight. At the architectural level, the project converges on a multi-head decoder in which a compact head produces the shortlist, a rerank head contributes shortlist-local discrimination, and a rich head preserves a broader latent representation suitable for regression and qualitative generation. At the single-model level, the strongest individual retrieval expert is strengthened by Monte Carlo test-time augmentation on a 197K token-space model, which reduces per-trial prediction variance at $16\times$ inference-time compute. At the final-system level, the headline retrieval result is obtained by a frozen cross-architecture score fusion that combines this MC-TTA-augmented expert with two complementary 768-D experts. The exported system reaches 86.3% CSLS Recall@1 on SHARED1000, while the earlier V35 plus N1v28a fixed fusion at 77.2% remains an important methodological milestone.

Together these results support the thesis hypothesis that reliable decoding in this regime depends not only on latent prediction quality, but also on how complementary experts are combined. The strongest generalisation evidence is held-out SHARED1000 retrieval; the manifold-alignment analyses of Section 5.5 are computed on the V62a single expert rather than on the fused system and are therefore presented as auxiliary evidence that decoded fMRI-to-CLIP embeddings preserve meaningful CLIP-space semantic geometry, not as a direct characterisation of the final triple-fusion endpoint. The fusion itself operates at score level rather than as an end-to-end fusion-trained architecture, and therefore improves retrieval without producing a unified fused latent embedding.

From a scientific perspective, one of the most important outcomes is interpretive clarity. The thesis does not claim that the problem is solved, nor that the current system matches the strongest large-scale shared-subject methods in the literature. Instead, it narrows the explanation of the remaining gap. Early obstacles were caused by concrete bugs and unstable optimization regimes. After those were removed, the dominant limitations became structural: limited scale, imperfect alignment, and the nontrivial geometry of high-dimensional retrieval spaces.

The thesis also makes an original methodological contribution by converting many negative results into useful knowledge. Several plausible ideas failed to improve performance in practice, and documenting those failures is scientifically valuable because it reduces future search space and helps distinguish between cosmetic innovation and genuinely effective design choices. Equally important, the implementation chapter shows that a software-intensive research project can only support valid scientific claims if analysis, design, testing, and reproducibility are handled explicitly.

There are several clear directions for future work:

- integrating stronger shared-subject learning and functional alignment, in the spirit of recent large-scale systems [13];
- extending the final retrieval formulation with fusion-aware training or fusion-aware checkpoint selection rather than relying only on post-hoc score combination;
- expanding the reconstruction analysis with quantitative image-generation metrics, curated qualitative panels, and a larger evaluation set beyond the current 16-example subset, and exploring stronger image-prompt conditioning such as IP-Adapter, which was deliberately scoped out of the current diffusion add-on;
- exploring whether richer token-level targets or ROI-aware encoders become more useful once scale and subject alignment improve;
- evaluating whether the final retrieval-oriented design can serve as a stable front-end for future perception-versus-imagery decoding experiments.

In conclusion, the thesis provides a reproducible, technically mature, and scientifically grounded step toward reliable fMRI-to-image decoding. Its main value lies not only in the final metric values, but also in the clarity of the resulting design principles: compact shortlist retrieval, dedicated reranking, and rich regression supervision should be treated as distinct roles, and complementary retrieval experts can be profitably fused after training when they demonstrably capture different parts of the signal. The CLIP-conditioned diffusion add-on strengthens this conclusion rather than replacing it: even when generation is steered by the fMRI-predicted CLIP embedding with best-of-N candidate selection and uncertainty-aware strength, exact retrieval remains the most trustworthy endpoint and the reconstruction stage is best read as a trade-off between low-level fidelity and semantic steering rather than as a uniform improvement.

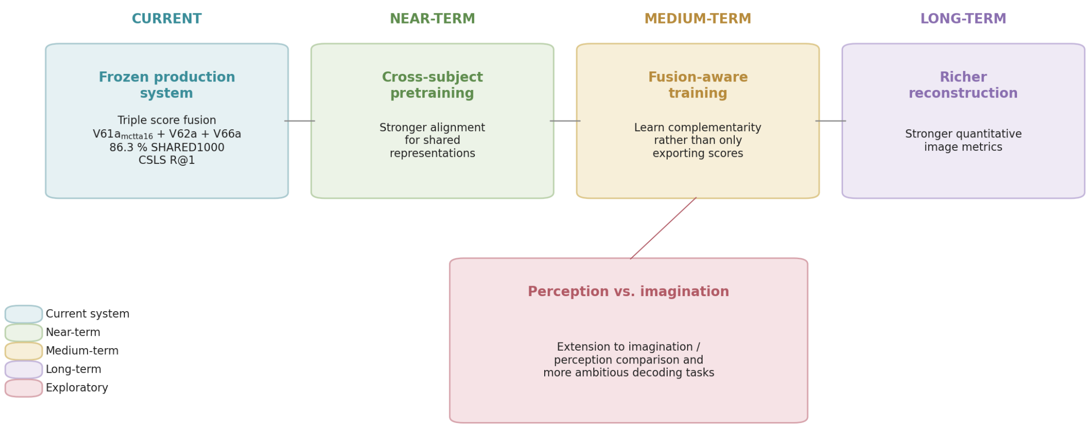


Figure 6.1: Roadmap after the final exported triple-fusion retrieval system. The next steps are not cosmetic tweaks to the current model, but deeper extensions: stronger shared-subject alignment, fusion-aware training, more rigorous reconstruction evaluation, and broader perception-versus-imagery decoding.

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